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INTERNATIONAL STANDARD

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Use case methodology –
Part 3: Definition of use case template artefacts into an XML serialized format

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

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Part 3: Definition of use case template artefacts into an XML serialized format

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CDV	Report on voting
SyCSmartEnergy/28/CDV	SyCSmartEnergy/48/RVC

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 62559 series, published under the general title Use case methodology, can be found on the IEC website.

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- withdrawn,
- replaced by a revised edition, or
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INTRODUCTION

For complex systems, the use case methodology supports a common understanding of functionalities, Actors and processes across different technical committees or even different organizations. Developed as software engineering tool, the methodology can be used to support the development of standards as it facilitates the analysis of requirements in relation to new or existing standards. Further arguments for the use case methodology and background information are available in IEC 62559-1.

Figure 1 provides an overview of the intended first parts of the IEC 62559 series, mainly describing the relation between IEC 62559-2 and IEC 62559-3.

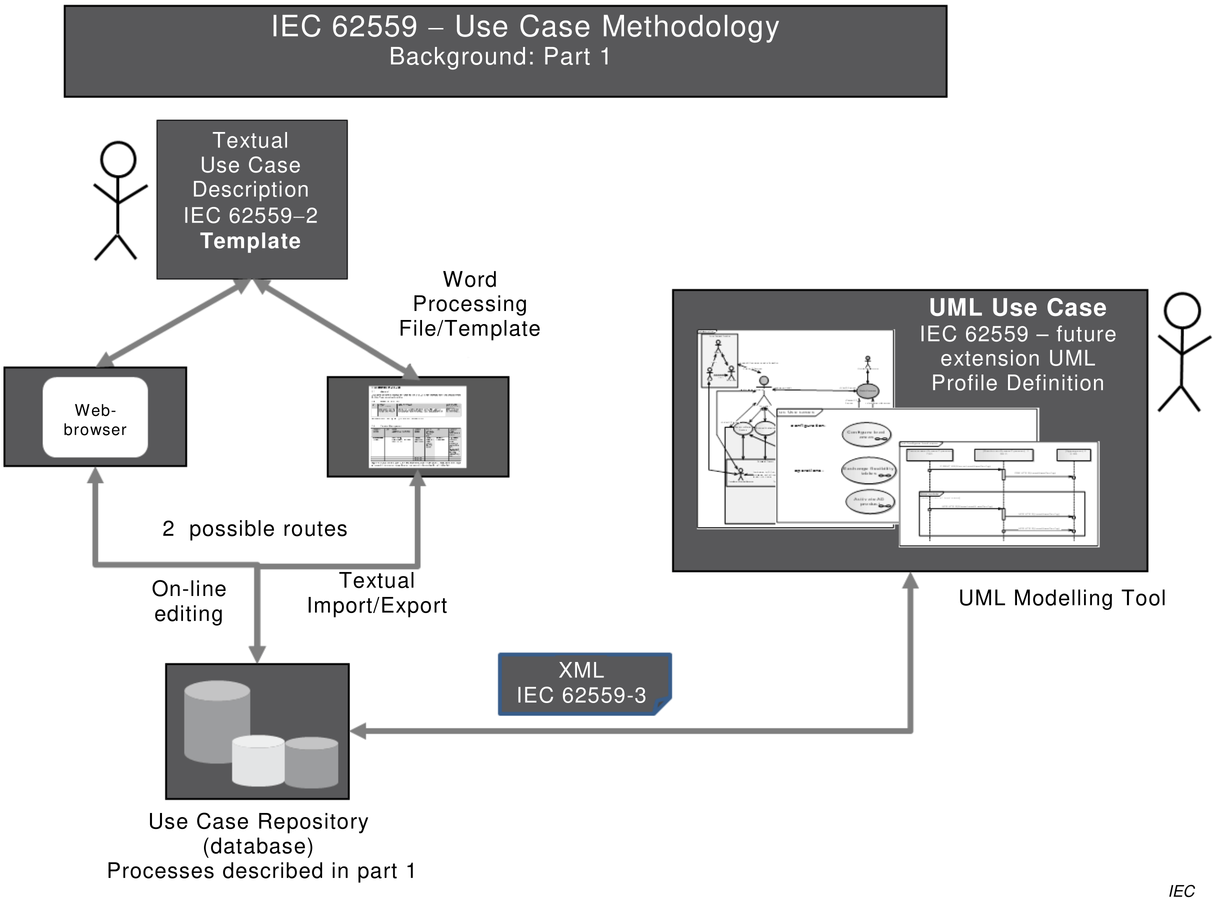

Figure 1 – IEC 62559 standard series

IEC 62559-1 – Concept and processes in standardization

IEC 62559-1 will be the basis for a common use case repository in order to gather use cases within IEC on a common collaborative platform. This repository will also be used to organize a harmonization of use cases in order to provide broadly accepted generic use cases as a basis for further standardization work. It describes processes and provides basics for the use case methodology like terms or use case types.

IEC 62559-2 – Definition of the templates for use cases, Actor list and requirements list

IEC 62559-2 defines the structure of a use case template, an Actor list and a list for requirements. The document is mainly based on IEC PAS 62559:2008 and shall be read together with IEC 62559-1.

IEC 62559-3 – Definition of use case template artefacts into an XML serialized format

Based on IEC 62559-2, this document defines the required core concepts and their serialization into an XML format of a use case template, an Actor list and a list for detailed requirements. The XML format is used to transfer the content of the template to other engineering systems (e.g. UML modelling tools). These documents are developed using the energy system and Smart Grids as examples, but they are general enough to be transferred to other domains and systems. It is intended to develop a UML profile definition based on this part in the future.

The IEC 62559 series is needed to fulfil the SG3 decision 7 made by the SMB at its February 2010 meeting (SMB/4204/DL, Decision 137/10) requesting the urgent delivery of a generic use case repository for all Smart Grid applications. Nevertheless, the use case methodology described in this document is intended for a broader application within standardization exceeding Smart Grid systems.

More and more complex systems such as Smart Grids or Smart Cities are raising the question of managing system level requirements, which have to be fed by many domains of expertise (in standardization related to different Technical Committees (TCs)), and which have to be broken down further and shared by many TCs in charge of specifying standards to support these system level functions.

One way to handle this transversality efficiently is to set some common methods and terms. The use case methodology is the current state of the art and supports further engineering activities.

The use case methodology offers a unique way for sharing ideas and requirements of new use cases or business cases between many experts/TCs with different backgrounds: for example, domain experts with knowledge about energy systems or business processes on one hand and system-/IT-experts defining exchanged information and communication on the other hand. In the requirement development process, domain experts provide general ideas and functional requirements. The main goal is for system experts to detail down these use cases to a level where they can be used to specify interfaces, dedicated functionality, data and service model exchange. However, safety- or EMC-experts (as examples) may also make use of the described use cases, their terminology and identified requirements.

However the starting point is to set up a framework for consistency within IEC, helping IEC members to provide use cases in a consistent manner – IEC 62559 serves as a basis for use case repositories in order to gather, administrate, maintain, and evaluate use cases.

Within IEC, a use case repository serves as common collaborative platform for use case elaboration and to organize a harmonization of use cases in order to provide broadly accepted generic use cases as a basis for further standardization work.

But the use case template defined in this document may serve not only for the development of standards, but also – as was the original purpose of IEC PAS 62559:2008 – as a helpful means for the realization of projects within the area of complex systems. Also other applications, which need the benefits of a structured requirements development and formalized description of functionality, may make use of the suggested template.

The use case methodology is seen as a process which starts with the definition of business ideas, goals and requirements, detailing these in use case descriptions. This information can be used as a basis to identify/link reference architectures describing the types of components used, and going further down to an analysis for the further standardization process.

Further developments regarding the use case template are expected. These developments are mainly related to information, which is required in the use case description for further analysis, and which can be mapped to other information (e.g. to a reference architecture, IT security methods, standards and data models). Partly this is considered in the suggested template of IEC 62559-2. Further relations will be described separately as they are still under development and they can be considered for the further development of the IEC use case repository.

USE CASE METHODOLOGY –

Part 3: Definition of use case template artefacts into an XML serialized format

1 Scope

In order to exchange use cases based on the template which is defined in IEC 62559-2, this part of IEC 62559 establishes the interfaces between the different use case repositories and/or UML engineering software tools.

Therefore, this document defines the required core concepts and their serialization into XML syntactic format of a use case template, an Actor list and list for detailed requirements.

As shown in Figure 2, the modelling approach is leveraging the use of UML in order to graphically represent the data contained in a use case based on the IEC 62559 template. Therefore the textual format of the use case template may be in the use case development process just a starting point for business experts or an easy way to modify use case data for non UML experts. As a consequence, it is important for the IEC 62559 series to provide a reliable way for converting this textual format into UML format and reciprocally. As soon as a use case repository is maintained based on the IEC 62559 series, another related need is to be able to import/export between different UML tools and different use case repositories the use case related information based on a tool independent format.

The main purpose of this document is to propose an independent format for transferring the use case information between modelling software. In order to satisfy this goal, the syntactic XML format is chosen to serialize the use case data. This document defines in detail the core concepts of the template into UML and their transformations into XML using the XSD standard.

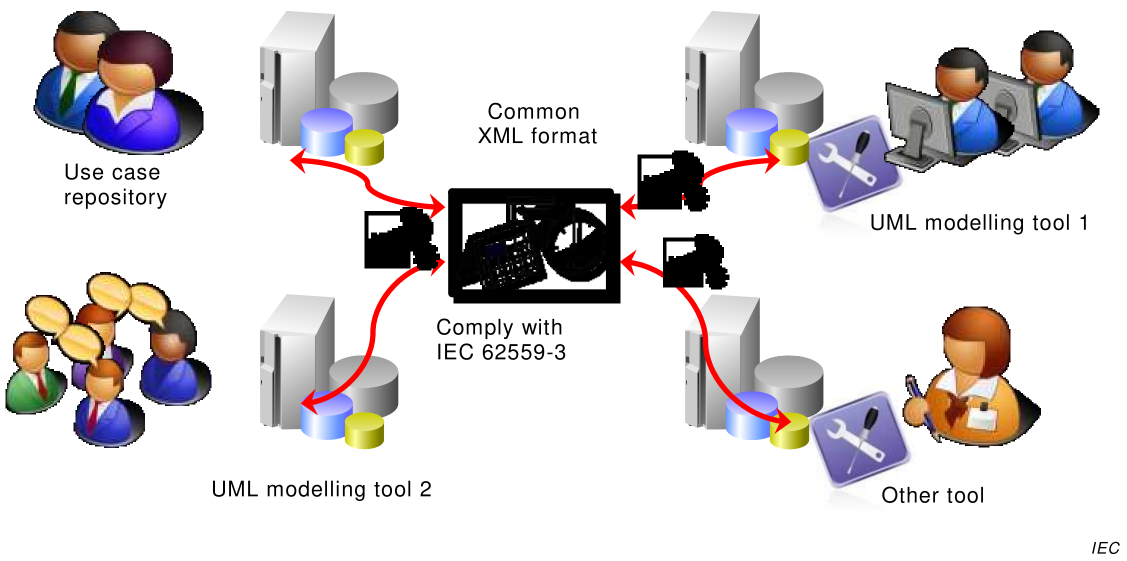

Figure 2 – A common XML format for importing/exporting use case information between a variety of modelling software and repositories

This document, as shown in Figure 2, supports the interoperability between modelling software used by different companies over the world.

Once this level of interoperability is achieved, IEC 62559 can provide a reliable mechanism to interpret those XML data in order to represent graphically UML use cases. This need will be covered as well in a future part of IEC 62559 (to be defined).

This document focuses on a methodological framework which is also used by IEC TC 57 standards and which is summarized in Clause 4.

In order to exchange use cases based on the template which is defined in IEC 62559-2, this document establishes the interfaces between the different use case repositories and/or UML tools.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 62559-2, *Use case methodology – Part 2: Definition of the templates for use cases, actor list and requirements list*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 62559-2 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

3.1
XML
eXtensible Markup Language
markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable

Note 1 to entry: The XML specification is produced by the World Wide Web Consortium (W3C).

3.2
XSD
XML Schema Definition Language
language used to define the structure, content, and semantics of XML files

Note 1 to entry: XSD is a recommendation by the World Wide Web Consortium (W3C).

4 Methodological framework for developing this standard

4.1 Model-based approach for developing the data exchange format

This document relies on a model-based approach for the definition of exchange formats and therefore defines respective UML models. The approach utilized herein has been developed in IEC TC 57 to develop specific data models (profiles) and is actively used for developing the Common Information Model (CIM) standards of IEC TC 57 for the electricity system communications. Within the approach several types of models, their interrelationships, and rules to derive more specific models from a generic model are described. As the approach is explained in detail in IEC 62325-450, 4.1 only briefly describes the approach and its adaptions to create the framework on which this document is built.

Using the approach enables several XML-based exchange formats to be defined for a particular subject. However, all exchange formats created according to this process are based on a single source model ensuring semantic compatibility. Two interrelated model types are used for different purposes, which play a key role in this process, labelled as Information Model and Contextual Model, respectively. Figure 3 outlines these models in the context of the data exchange format definition process.

The Information Model represents all common core concepts of a use case that are identified in IEC 62559-2 on a rather generic level, including entities, their attributes, and relations. This comprises for instance a use case itself, scenarios of a use case, involved Actors, or information objects exchanged throughout a use case. Concepts defined in the Information Model define the semantics of the data exchange. Consequently, the Information Model provides a “single point of truth” regarding the structuring of use case information and can be seen as strictly normative.

A Contextual Model is always based on the Information Model. Contextual Models are created to cover specific needs of information exchange with regard to the Information Model. Contextual Models include a context-dependent subset of (or even the complete) Information Model that can be further constrained or specialized.

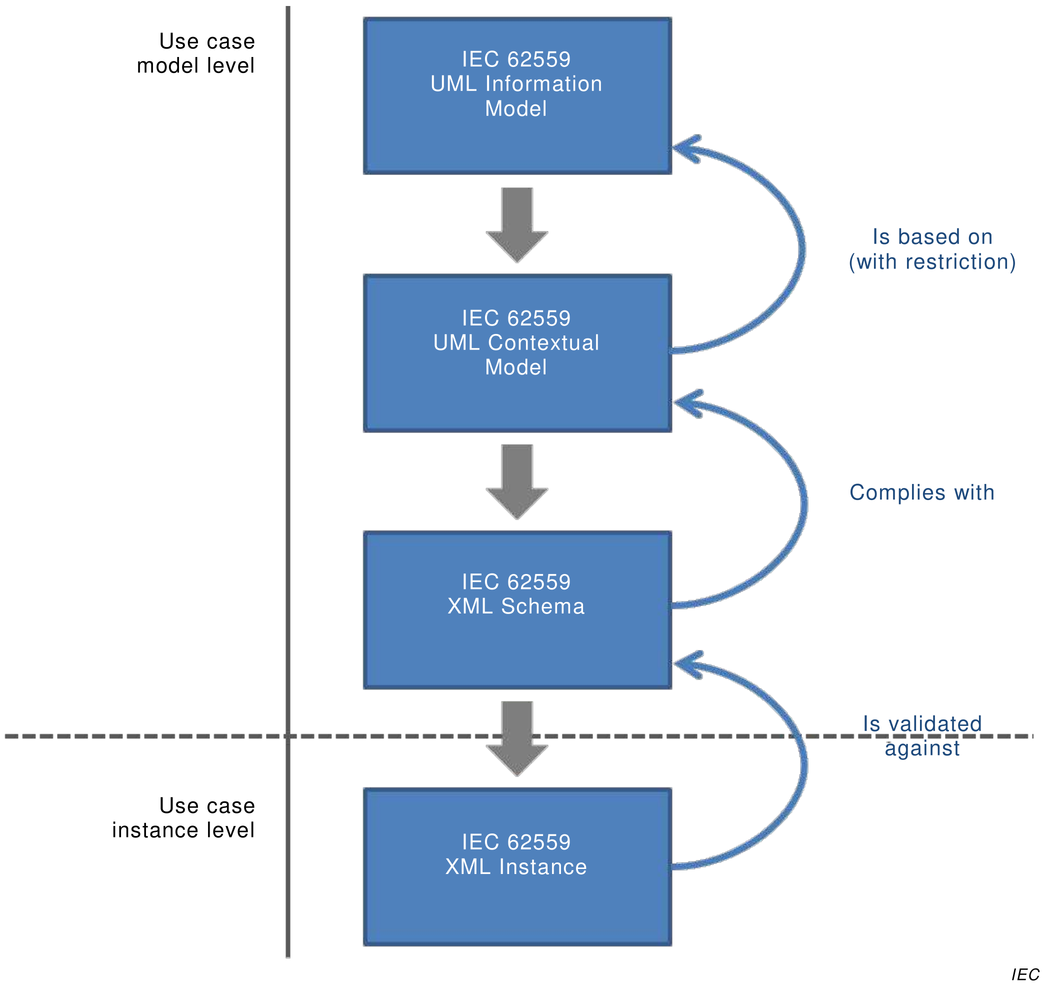

Figure 3 – Model-based development framework for the definition of the IEC 62559-3 standard XML-based data exchange format

Finally further rules can be applied to a Contextual Model creating the final exchange format that then inherently complies with the Contextual Model. In the context of this document, this is an XML schema that serves as the basis for a validation of exchanged use case data. For this purpose, use case instance data shall be defined in an XML document.

The model-based process outlined above was applied to develop respective XML-based data exchange formats. In order to finally produce a standardized XML schema definition (XSD) file according to IEC 62559-2, the model-based approach is carried out in detail as follows.

a) IEC 62559 UML Information Model:

   This UML gathers the core concepts described in IEC 62559-2. This is modelled using UML class diagrams including classes, attributes, data types, and associations. This Information Model contains the core concepts of a use case modelled in a generic way in order to fit other reuses of those concepts in other implementations. For example, this Information Model could also be used to define the structure of a data base for use cases. Therefore, this Information Model is made flexible by modelling concepts with optional aspects. For example, many attributes and associations are modelled with flexible cardinalities such as 0..1 for attributes and 0..* for relationships. In order to create an implementation of this generic Information Model, it is necessary to define more precisely how the core concepts are used for the specific purpose of the exchange of use case information.

b) IEC 62559 UML Contextual Model:

This second UML model is based on the UML Information Model by modelling restrictions in order to adapt it for the specific purpose of defining a model specifying the exchange of use case information. Those restrictions set on the Information Model concepts make this Contextual Model accurate for the exchange of use case information. The allowed restrictions are the following ones:

1) Restrictions on cardinalities for attributes and associations by narrowing them. For example, an attribute cardinality 0..1 can be set as 1..1 (mandatory attribute) or 0..1 (optional attribute) or 0..0 (not used attribute). An association end cardinality 0..* can be set to an included interval such as 2..6 (sequence of instanced concepts at minimum 2 and maximum 6).
2) Restrictions on attribute value space by adding facets. For example, an attribute typed by a String can have a facet of length 6, meaning that the expected attribute value is a sequence of 6 symbols.

c) IEC 62559 XML schema:

This XML schema definition (XSD) is automatically generated from the previously developed UML Contextual Model. Every XML use case information exchange shall be validated against this XSD. It thus ensures interoperability between modelling software because the use case information exchange must respect the structure defined in the XSD.

4.2 Example for the model-based process in the context of IEC 62559

Subclause 4.2 illustrates an example based on the relationship between a use case scenario and an Actor playing the PrimaryActor role in this scenario.

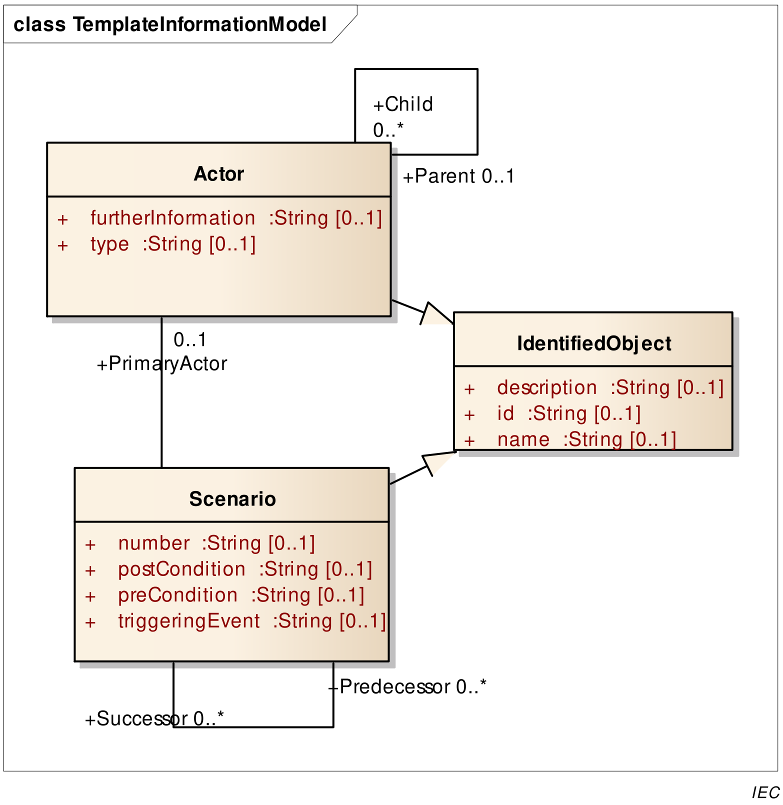

Figure 4 – UML Information Model for the Scenario-Actor relationship example

Figure 4 shows the UML Information Model representing the two core concepts Scenario and Actor (using UML classes) linked with a UML association having the role PrimaryActor and 0..1 cardinality on the Actor side. It means that a scenario can have an optional primary Actor. Supplementary information for both scenario and Actor is modelled with attributes of those two classes. For example, a scenario has a number, a postCondition, a precondition and a triggeringEvent. Because those two concepts have common needs for representing an identifier, a generalization class IdentifedObject shares those common attributes such as the id, name and description attributes.

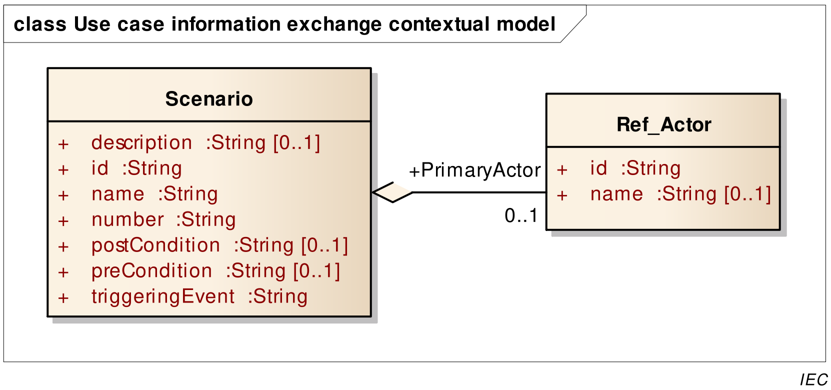

Figure 5 – UML Contextual Model based on the Scenario-Actor relationship example

In the UML Contextual Model (see Figure 5), the two concepts Scenario and Actor are reused and customized by adding restrictions on the Information Model for the specific purpose of exchanging information about a Scenario having a PrimaryActor. In this example, we make some attributes mandatory. Thus for Scenario, the identifier (id), the name, the number and triggeringEvent are mandatory information for the exchange. Moreover, if the information of a PrimaryActor is given, then its identifier is also mandatory. The concept Actor is reused several times with different defined structures in the same Contextual Model. In order to have a unique contextual name definition for those different structures, a qualifier is added to the Information Model class name. In this example, the contextual class based on the Actor concept is named Ref_Actor. When building the Contextual Model, the association between Scenario and Actor is restricted and becomes an aggregation, meaning that in the exchange of data, the scenario information includes information about a PrimaryActor.

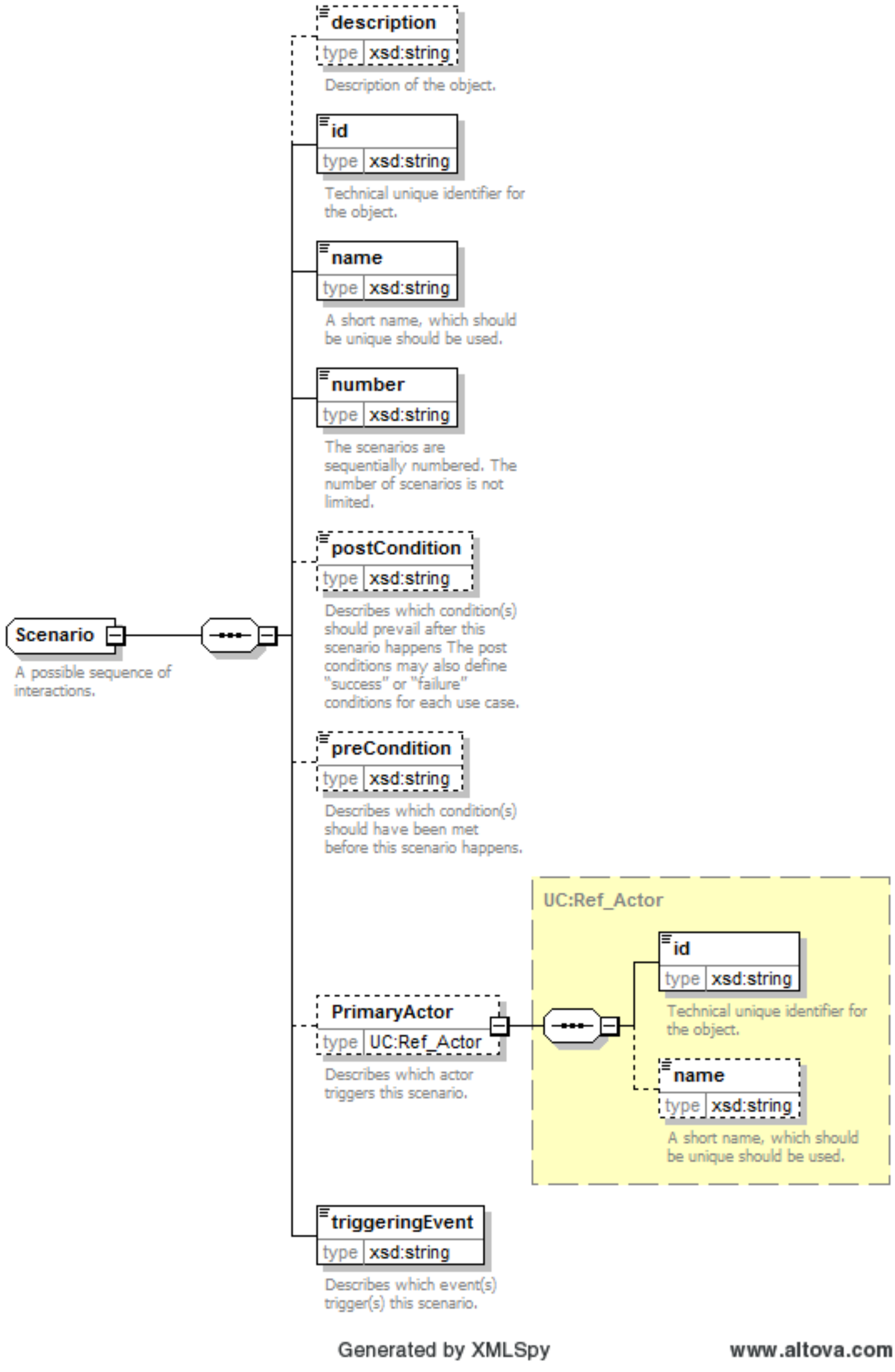

Figure 6 – XML Schema corresponding to the Scenario-Actor relationship example

Figure 6 illustrates graphically the XSD code which is automatically generated from the UML Contextual Model shown in Figure 5.

4.3 Development of specific data exchange profiles

Applying the model-based process described above can result in multiple Contextual Models and accordingly multiple different exchange formats, though usually with the same semantics. This process of deriving a specific model, in terms of a subset of model elements, from a generic model and specializing model elements, is referred to as profiling; the result of the profiling process is called a profile. As multiple profiles may be created based on a single Information Model, not all profiles can be considered normative. However, when data between two applications shall be successfully exchanged, there needs to be a (normative) agreement on a specific exchange format, not only semantically, but also syntactically.

Therefore, this document identifies key scenarios in regard to information exchange resulting in respective profiles that are normative and can be implemented in software. Key information exchange scenarios currently identified include the exchange of all concepts of the Information Model (e.g. for exchanging information between use case repositories), the exchange of a set of Actors (role model), and the exchange of requirements (see Clause 5). While the first data exchange scenario is rather generic, the latter ones enable a more thorough validation of the data in comparison to the exchange model covering an entire repository, as certain restrictions can be assumed, like the presence of an Actor or a requirement.

The normative data exchange format of the first generic exchange scenario can be found in Clause 6. Future editions of IEC 62559-3 may add further standardized information exchange scenarios and respective profiles, like for use cases. In case of specific needs, however, non-normative exchange formats/profiles can be developed using the model-based process.

5 Application of the methodological framework for defining a set of key syntactic exchange formats (use cases, actors, requirements)

With respect to different types of data exchange depending on specific key data exchange scenarios, IEC 62559-3 is intended to include a set of key data exchange formats for use cases, Actors, and requirements in particular. In order to produce those XML syntactic exchange formats, the modelling process described in Clause 4 is applied for designing each exchange format. Figure 7 provides an overview of the profiling of the generic Information Model.

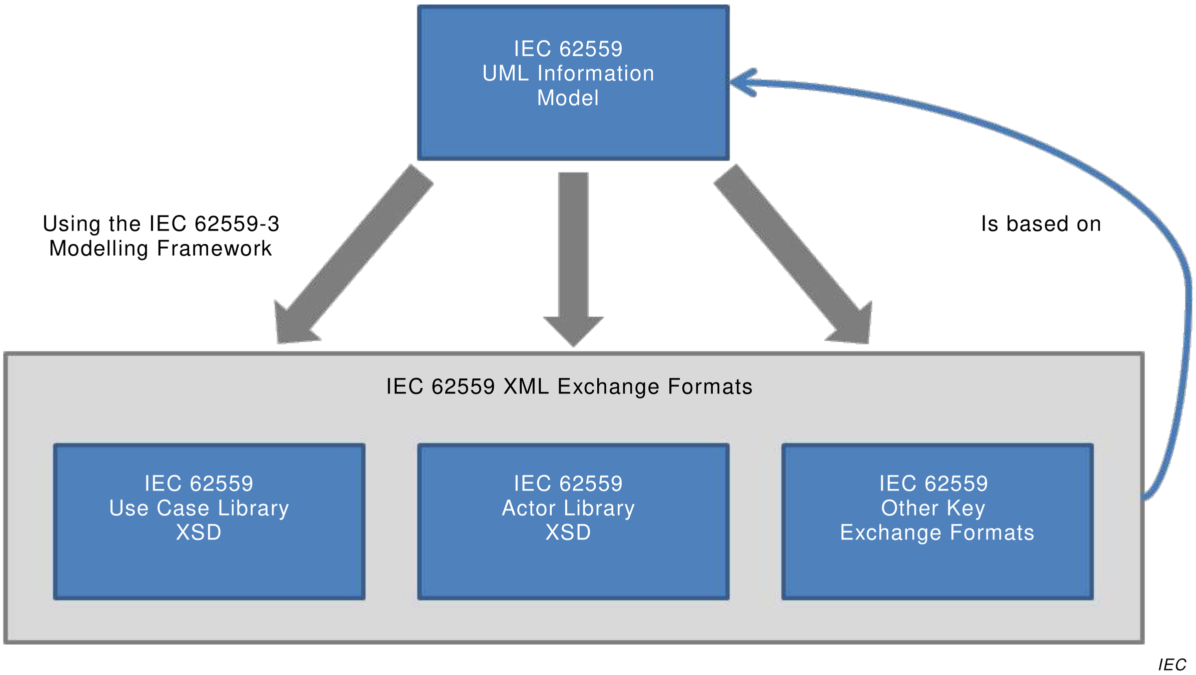

Figure 7 – Producing key XML exchange formats based on a unique IEC 62559 UML Information Model

6 Models' detailed content (automatic generation from UML models)

6.1 UML use case core concepts (UML Information Model)

6.1.1 General

The use case core concepts are described through a UML Information Model which is shared among the various defined exchange formats in this document. The core concepts are classified into a set of UML packages.

6.1.2 Model package UseCaseCoreConcepts

6.1.2.1 General

IEC 62559 UML Information Model for the use case core concepts.

Figure 8 shows class diagram UseCaseCore_Libraries.

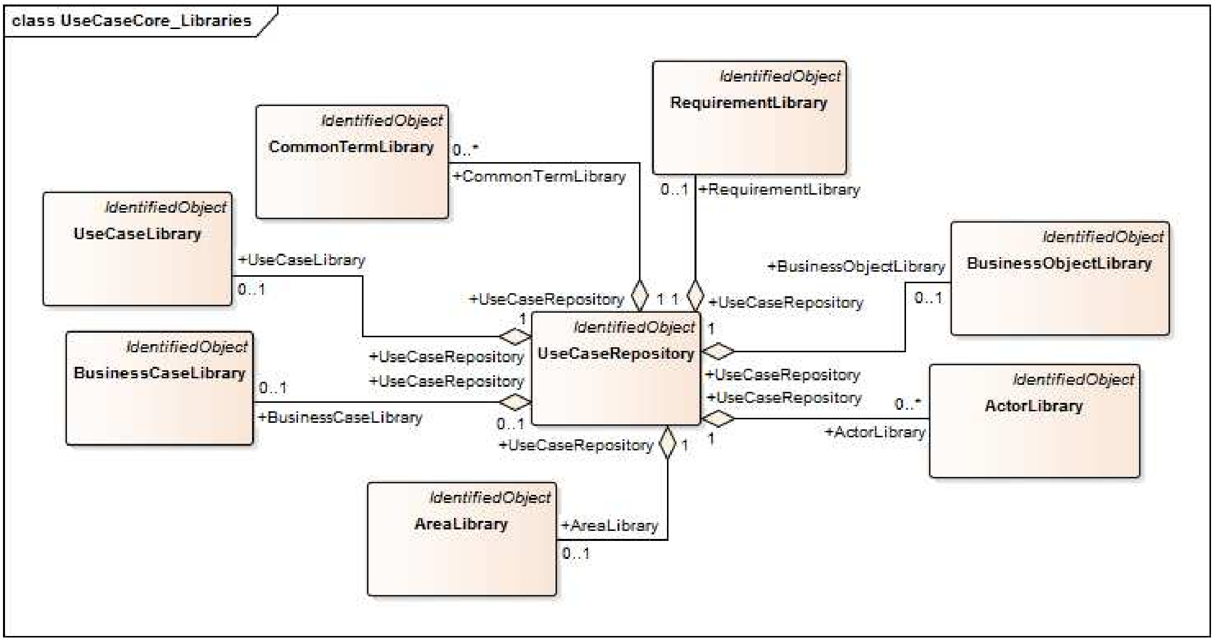

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Figure 8 – Class diagram UseCaseCoreConcepts::UseCaseCore_Libraries

Figure 9 shows class diagram UseCaseCore_ActorLibrary.

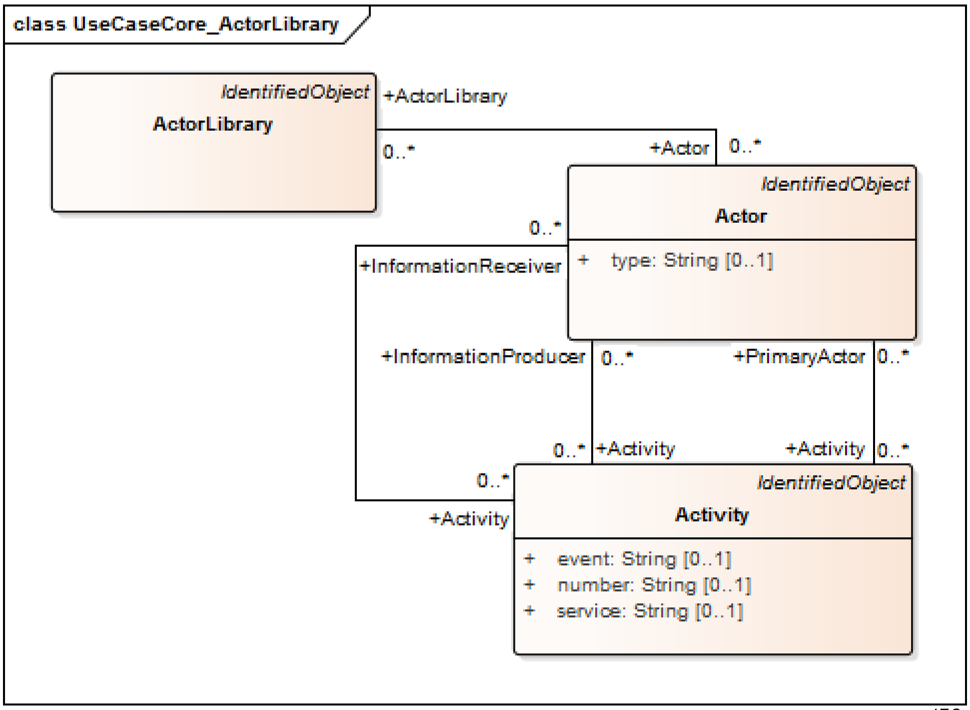

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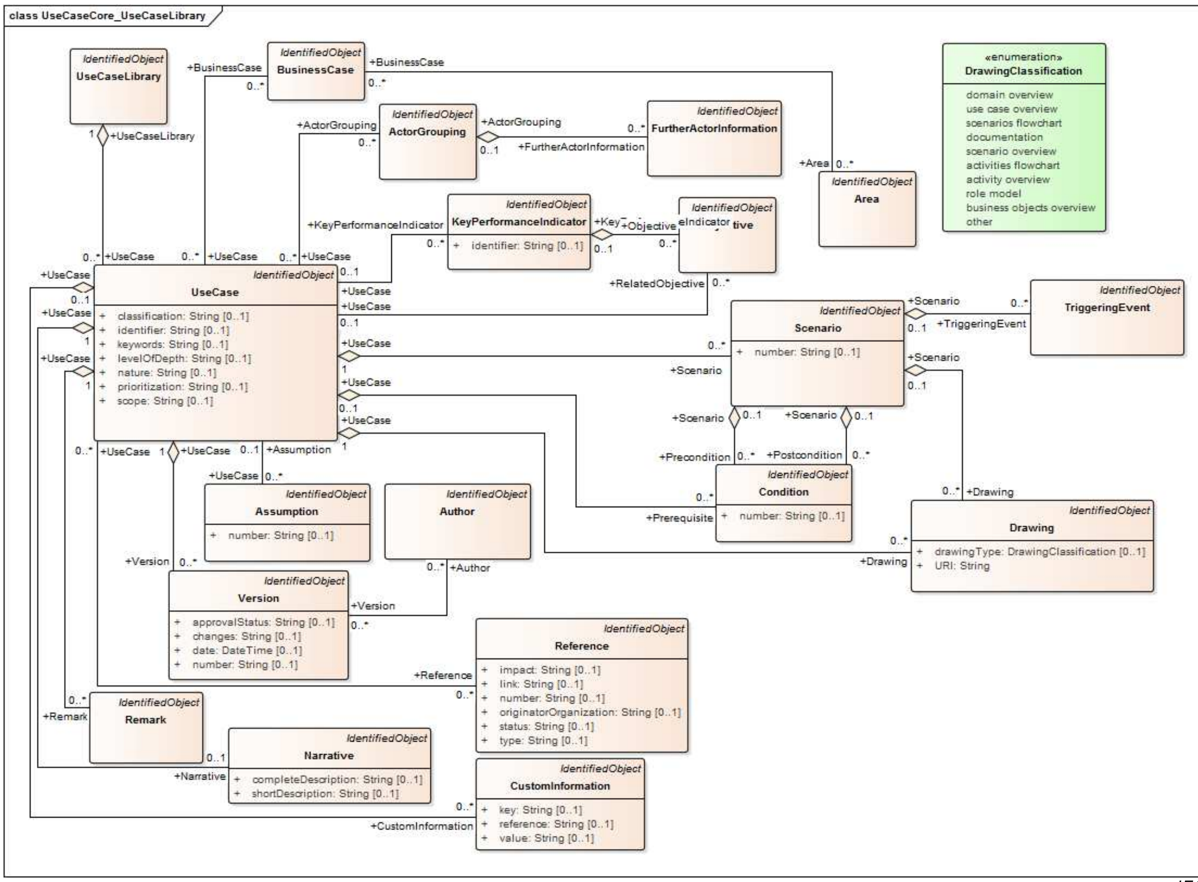
Figure 9 – Class diagram UseCaseCoreConcepts::UseCaseCore_ActorLibrary

Figure 10 shows class diagram UseCaseCore_UseCaseLibrary.



IEC

Figure 10 – Class diagram UseCaseCoreConcepts::UseCaseCore_UseCaseLibrary

Figure 11 shows class diagram UseCaseCore_DetailedActivities.

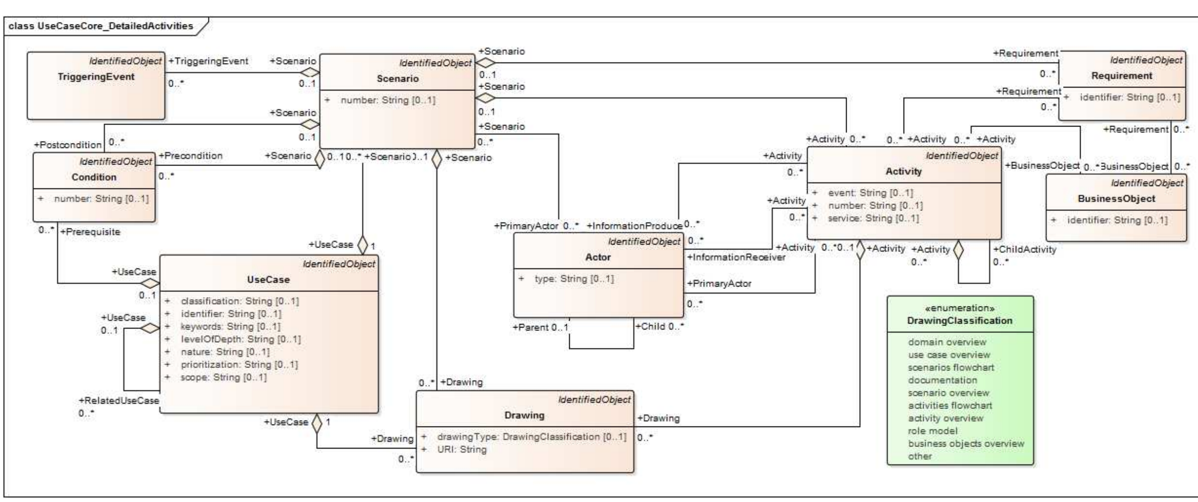


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Figure 11 – Class diagram UseCaseCoreConcepts::UseCaseCore_DetailedActivities

Figure 12 shows class diagram UseCaseCore_AreaLibrary.

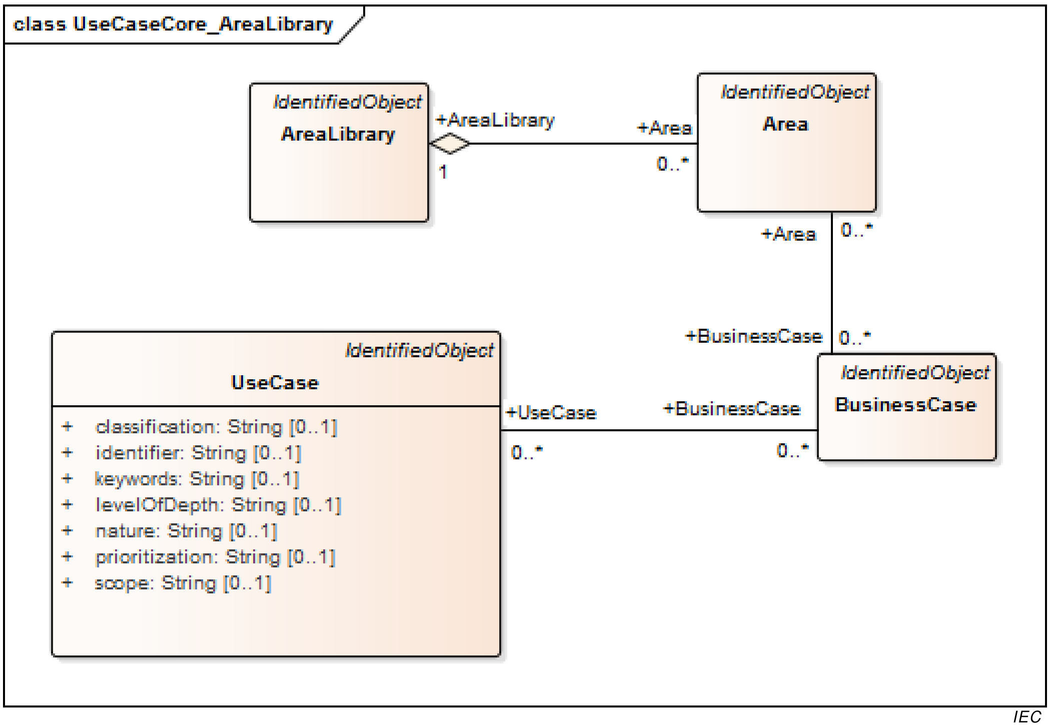

Figure 12 – Class diagram UseCaseCoreConcepts::UseCaseCore_AreaLibrary

Figure 13 shows class diagram UseCaseCore_BusinessCaseLibrary.

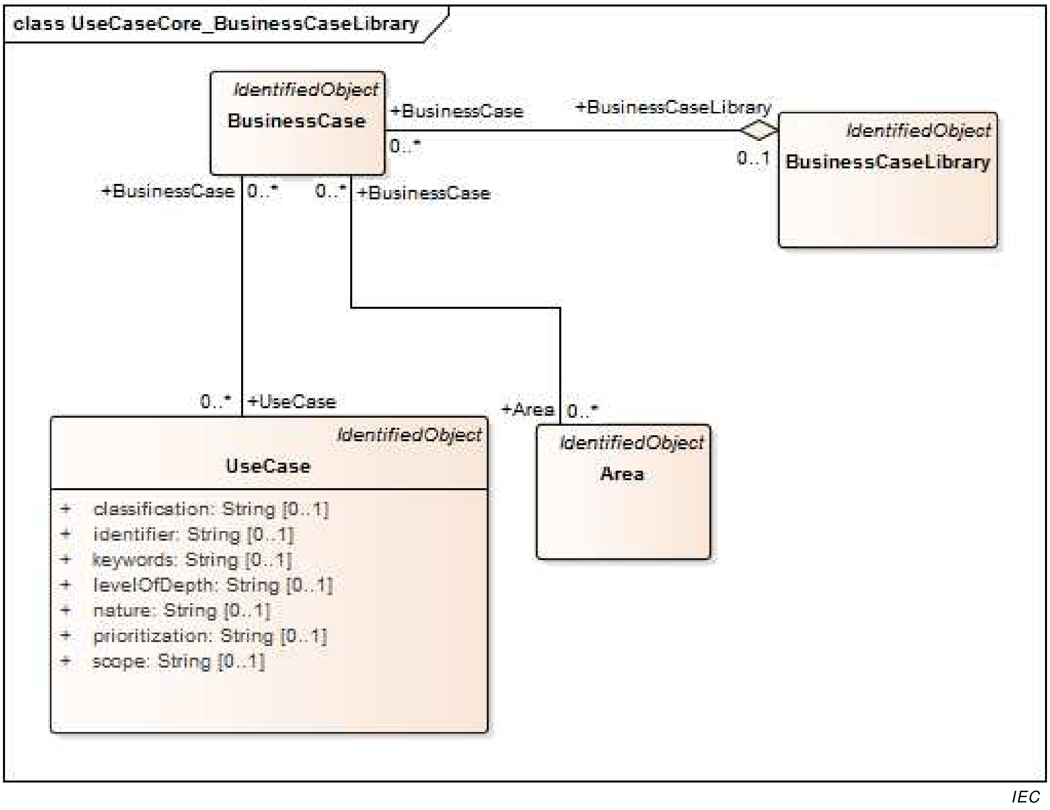

Figure 13 – Class diagram UseCaseCoreConcepts::UseCaseCore_BusinessCaseLibrary

Figure 14 shows class diagram UseCaseCore_BusinessObjectLibrary.

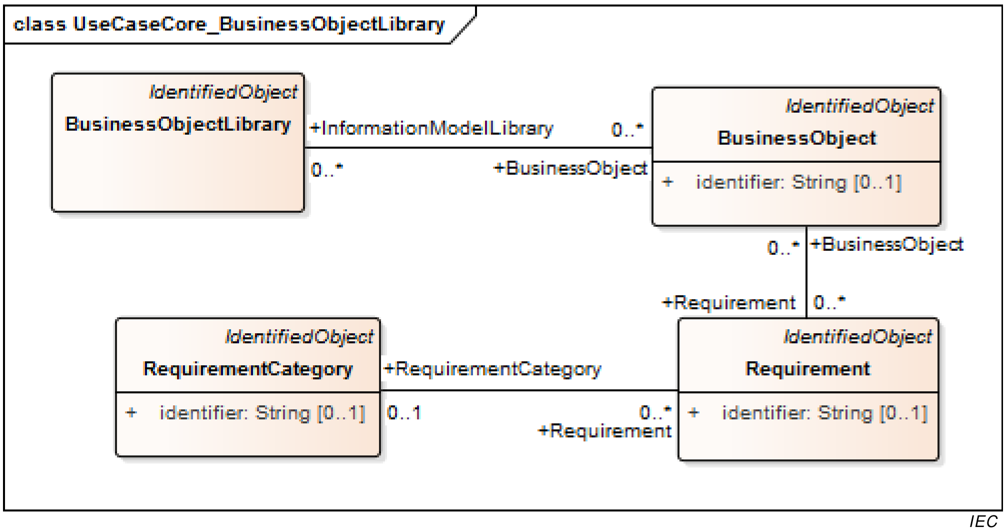

Figure 14 – Class diagram UseCaseCoreConcepts::UseCaseCore_BusinessObjectLibrary

Figure 15 shows class diagram UseCaseCore_CommonTermLibrary.

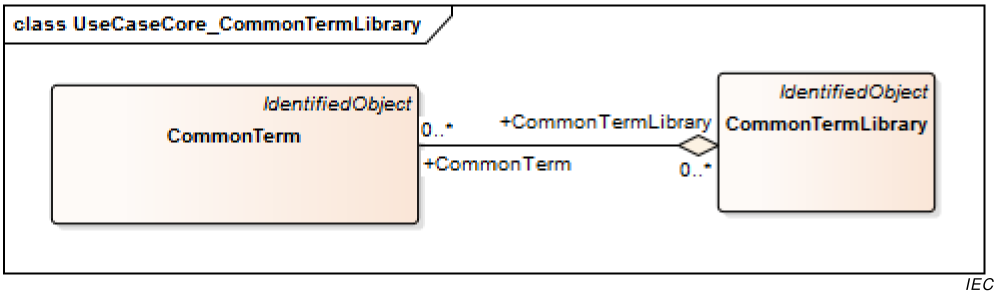

Figure 15 – Class diagram UseCaseCoreConcepts::UseCaseCore_CommonTermLibrary

Figure 16 shows class diagram UseCaseCore_RequirementLibrary.

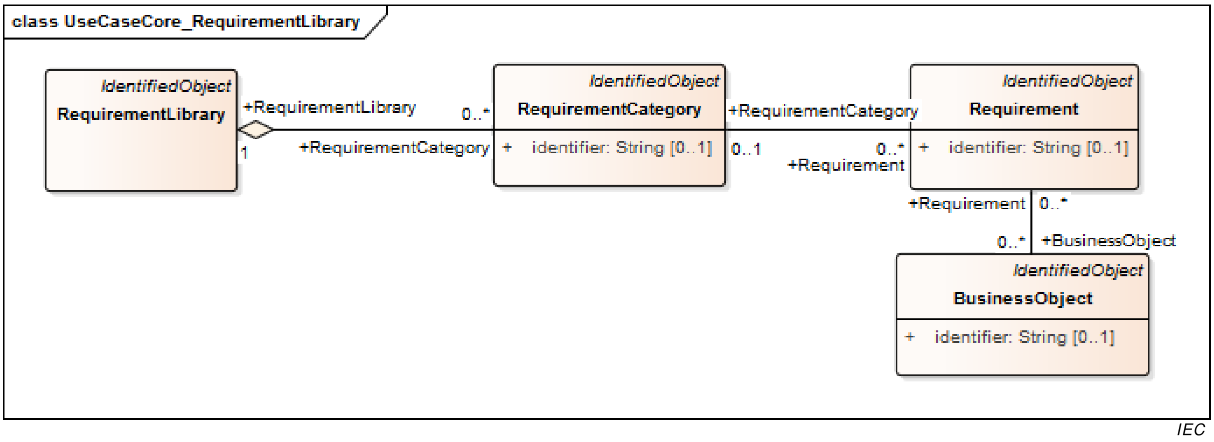

Figure 16 – Class diagram UseCaseCoreConcepts::UseCaseCore_RequirementLibrary

6.1.2.2 Activity

It may be subdivided into other activities or represent an elementary step of the scenario.

Table 1 shows all attributes of Activity.

Table 1 – Attributes of UseCaseCoreConcepts::Activity

name	type	description
event	String	It is the event that triggers the step. This might be completion of previous steps (in that case the event is the comma-separated list of their numbers) or the condition to exit a loop.
number	String	index of the activity
service	String	This column identifies the nature of the information flow and the originator of the information. Available options are CREATE, GET, CHANGE, DELETE, CANCEL, EXECUTE derived from IEC 61968-100:2013. Additionally, REPORT, TIMER and REPEAT are suggested. – CREATE means that an information object is to be created at the Producer. – GET (this is the default value if none is populated) means that the Receiver requests information from the Producer (default). – CHANGE means that information is to be updated. Producer updates the Receiver's information. – DELETE means that information is to be deleted. Producer deletes information from the Receiver. – CANCEL, CLOSE imply actions related to processes, such as the closure of a work order or the cancellation of a control request. – EXECUTE is used when a complex transaction is being conveyed using a service, which potentially contains more than one verb. – REPORT is used to represent transferral of unsolicited information or asynchronous information flows. Producer provides information to the Receiver. – TIMER is used to represent a waiting period. When using the TIMER service, the Information Producer and Information Receiver fields shall refer to the same Actor. – REPEAT is used to indicate that a series of steps is repeated until a condition or trigger event. The condition is specified as the text in the "Event" column for this row or step. Following the word REPEAT, the first and last step numbers of the series to be repeated shall appear, in parenthesis, in the following form: REPEAT(X-Y) where X is the first step and Y is the last step. These common service definitions are related to automation/information or communication systems. In case the use case template is applied in other domains, further services might be used and described.
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 2 shows all association ends of Activity with other classes.

Table 2 – Association ends of UseCaseCoreConcepts::Activity with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] Requirement	Requirement	Several requirements may be listed in one step, comma separated.
[0..*]	[0..*] InformationReceiver	Actor	This identifies the receiver of the information. This should be one of the Actors defined in the repository.
[0..*]	[0..*] InformationProducer	Actor	This identifies the producer or source of the information. This should be one of the Actors defined in the repository.
[0..*]	[0..*] BusinessObject	BusinessObject	This describes briefly the information to be exchanged between the two Actors – information producer and information receiver. It may list several information items in one step, comma separated.
[0..1]	[0..*] Drawing	Drawing	the list of associated drawing
[0..*]	[0..*] PrimaryActor	Actor	reference to the PrimaryActor
[0..*]	[0..*] ChildActivity	Activity	child activities
[0..*]	[0..1] Scenario	Scenario	reference to the scenario
[0..*]	[0..*] Activity	Activity	parent activities

6.1.2.3 Actor

It is the entity that communicates and interacts.

These Actors can include people, software applications, systems, databases, and even the power system itself.

Table 3 shows all attributes of Actor.

Table 3 – Attributes of UseCaseCoreConcepts::Actor

name	type	description
type	String	type of the Actor (business, system, ...)
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 4 shows all association ends of Actor with other classes.

Table 4 – Association ends of UseCaseCoreConcepts::Actor with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..*] Child	Actor	list of specialized Actors
[0..*]	[0..*] Activity	Activity	the activities related to the Actor
[0..*]	[0..*] Activity	Activity	the activities related to this Actor
[0..*]	[0..*] Activity	Activity	An Actor can be linked to an activity.
[0..*]	[0..*] Scenario	Scenario	The Actor is linked to scenarios.
[0..*]	[0..*] ActorLibrary	ActorLibrary	The Actor is referenced by libraries.
[0..*]	[0..*] UseCase	UseCase	list of use cases in relation with an Actor
[0..*]	[0..*] FurtherActorInformation	FurtherActorInformation	An Actor may have a list of further information shared between different groupings for different use cases.
[0..*]	[0..1] Parent	Actor	the generalized Actor

6.1.2.4 ActorGrouping

Group of Actors used to organize an Actor list

Table 5 shows all attributes of ActorGrouping.

Table 5 – Attributes of UseCaseCoreConcepts::ActorGrouping

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 6 shows all association ends of ActorGrouping with other classes.

Table 6 – Association ends of UseCaseCoreConcepts::ActorGrouping with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] UseCase	UseCase	ActorGrouping is associated with several use cases.
[0..1]	[0..*] FurtherActorInformation	FurtherActorInformation	In a context of one use case, an Actor grouped into an ActorGrouping may have one FurtherActorInformation.

6.1.2.5 ActorLibrary

Actors of the repository

Table 7 shows all attributes of ActorLibrary.

Table 7 – Attributes of UseCaseCoreConcepts::ActorLibrary

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 8 shows all association ends of ActorLibrary with other classes.

Table 8 – Association ends of UseCaseCoreConcepts::ActorLibrary with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] Actor	Actor	the Actors listed in the library
[0..*]	[1..1] UseCaseRepository	UseCaseRepository	the associated use case repository

6.1.2.6 Area

Area of knowledge or activity characterized by a set of concepts and terminology understood by the practitioners in that area

Table 9 shows all attributes of Area.

Table 9 – Attributes of UseCaseCoreConcepts::Area

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 10 shows all association ends of Area with other classes.

Table 10 – Association ends of UseCaseCoreConcepts::Area with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1] ParentArea	Area	link with a parent area
[0..*]	[1..1] AreaLibrary	AreaLibrary	the library linking the area
[0..*]	[0..*] BusinessCase	BusinessCase	related business cases associated with the domain
[0..1]	[0..*] SubArea	Area	link with a sub area

6.1.2.7 AreaLibrary

Areas of the repository

Table 11 shows all attributes of AreaLibrary.

Table 11 – Attributes of UseCaseCoreConcepts::AreaLibrary

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 12 shows all association ends of AreaLibrary with other classes.

Table 12 – Association ends of UseCaseCoreConcepts::AreaLibrary with other classes

[mult from]	[mult to] name	type	description
[1..1]	[0..*] Area	Area	the list of areas within the library
[0..1]	[1..1] UseCaseRepository	UseCaseRepository	the associated repository

6.1.2.8 Assumption

It may be used to define a further general assumption for a use case, such as which systems already exist, which contractual relations exist, and which configurations of systems are probably in place. Initial states of information exchanged shall also be identified.

Table 13 shows all attributes of Assumption.

Table 13 – Attributes of UseCaseCoreConcepts::Assumption

name	type	description
number	String	index of the assumption
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 14 shows all association ends of Assumption with other classes.

Table 14 – Association ends of UseCaseCoreConcepts::Assumption with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1] UseCase	UseCase	the link toward the related use case

6.1.2.9 Author

This field is used to document who has provided the current version. It can be a person, organization or, for example, standardization committee like a technical committee (TC) or working group (WG).

Table 15 shows all attributes of Author.

Table 15 – Attributes of UseCaseCoreConcepts::Author

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 16 shows all association ends of Author with other classes.

Table 16 – Association ends of UseCaseCoreConcepts::Author with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] Version	Version	the related versions in which the author has participated

6.1.2.10 BusinessCase

It provides a description or reference with some rationale for the suggested use case. Usually the business case is related to several use cases.

Table 17 shows all attributes of BusinessCase.

Table 17 – Attributes of UseCaseCoreConcepts::BusinessCase

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 18 shows all association ends of BusinessCase with other classes.

Table 18 – Association ends of UseCaseCoreConcepts::BusinessCase with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] Area	Area	Use cases can be used in various areas (e.g. energy system). Within these areas different domains are used to define / determine a more specific subgrouping. One or more domains and zones, comma separated, can be specified in the template.
[0..*]	[0..*] UseCase	UseCase	list of use cases covering the business case
[0..*]	[0..1] BusinessCaseLibrary	BusinessCaseLibrary	the library containing the business case

6.1.2.11 BusinessCaseLibrary

Business cases of the repository

Table 19 shows all attributes of BusinessCaseLibrary.

Table 19 – Attributes of UseCaseCoreConcepts::BusinessCaseLibrary

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 20 shows all association ends of BusinessCaseLibrary with other classes.

Table 20 – Association ends of UseCaseCoreConcepts::BusinessCaseLibrary with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..*] BusinessCase	BusinessCase	the list of business cases
[0..1]	[0..1] UseCaseRepository	UseCaseRepository	the use case repository associated with the library

6.1.2.12 BusinessObject

Business object exchanged in detailed activity steps

Table 21 shows all attributes of BusinessObject.

Table 21 – Attributes of UseCaseCoreConcepts::BusinessObject

name	type	description
identifier	String	identification used in references in the use case description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 22 shows all association ends of BusinessObject with other classes.

Table 22 – Association ends of UseCaseCoreConcepts::BusinessObject with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] Requirement	Requirement	It can be used to define requirements related to the information and not to the step as in the step by step analysis.
[0..*]	[0..*] Activity	Activity	the activity which uses this exchanged business object
[0..*]	[0..*] InformationModelLibrary	BusinessObjectLibrary	the associated library

6.1.2.13 BusinessObjectLibrary

Business objects which are exchanged in use cases

Table 23 shows all attributes of BusinessObjectLibrary.

Table 23 – Attributes of UseCaseCoreConcepts::BusinessObjectLibrary

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 24 shows all association ends of BusinessObjectLibrary with other classes.

Table 24 – Association ends of UseCaseCoreConcepts::BusinessObjectLibrary with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] BusinessObject	BusinessObject	list of business objects within the library
[0..1]	[1..1] UseCaseRepository	UseCaseRepository	the associated repository

6.1.2.14 CommonTerm

It represents a glossary term and its definition.

Table 25 shows all attributes of CommonTerm.

Table 25 – Attributes of UseCaseCoreConcepts::CommonTerm

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 26 shows all association ends of CommonTerm with other classes.

Table 26 – Association ends of UseCaseCoreConcepts::CommonTerm with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] UseCase	UseCase	the associated use case
[0..*]	[0..*] CommonTermLibrary	CommonTermLibrary	the associated library

6.1.2.15 CommonTermLibrary

Common glossary defining terms for all use cases

Table 27 shows all attributes of CommonTermLibrary.

Table 27 – Attributes of UseCaseCoreConcepts::CommonTermLibrary

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 28 shows all association ends of CommonTermLibrary with other classes.

Table 28 – Association ends of UseCaseCoreConcepts::CommonTermLibrary with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] CommonTerm	CommonTerm	list of common terms
[0..*]	[1..1] UseCaseRepository	UseCaseRepository	the associated repository

6.1.2.16 Condition

It describes either a precondition or a postcondition.

Table 29 shows all attributes of Condition.

Table 29 – Attributes of UseCaseCoreConcepts::Condition

name	type	description
number	String	index of the condition
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 30 shows all association ends of Condition with other classes.

Table 30 – Association ends of UseCaseCoreConcepts::Condition with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1] UseCase	UseCase	the associated Use case
[0..*]	[0..1] Scenario	Scenario	reference to the scenario
[0..*]	[0..1] Scenario	Scenario	reference to the scenario

6.1.2.17 CustomInformation

It provides a flexible option to include miscellaneous custom, semi-structured information which does not fit into other parts of the template.

Table 31 shows all attributes of CustomInformation.

Table 31 – Attributes of UseCaseCoreConcepts::CustomInformation

name	type	description
key	String	unique key for identification purposes
value	String	the corresponding information for the provided key
reference	String	This field refers to relevant template section.
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 32 shows all association ends of CustomInformation with other classes.

Table 32 – Association ends of UseCaseCoreConcepts::CustomInformation with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1] UseCase	UseCase	the associated use case

6.1.2.18 Drawing

For clarification, in general it is recommended to provide drawing(s) by hand, by a graphic or as UML graphics (preferred in IEC 62559). The drawing should show interactions which identify the steps where possible.

Table 33 shows all attributes of Drawing.

Table 33 – Attributes of UseCaseCoreConcepts::Drawing

name	type	description
drawingType	DrawingClassification	type of drawing
URI	String	resource Path to an image file ex: http://www..../image.jpg or path to an XMI file with a relative diagram path from the root of the UML model. ex: http://www.../Model.xmi#Package1/Package2/.../DiagName
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 34 shows all association ends of Drawing with other classes.

Table 34 – Association ends of UseCaseCoreConcepts::Drawing with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1] Activity	Activity	the associated activity
[0..*]	[1..1] UseCase	UseCase	the associated use case
[0..*]	[0..1] Scenario	Scenario	the associated scenario

6.1.2.19 DrawingClassification enumeration

Possible types of drawing

Table 35 shows all literals of DrawingClassification.

Table 35 – Literals of UseCaseCoreConcepts::DrawingClassification

literal	description
domain overview	overview of relationships between domains and use cases
use case overview	overview of use case short template information
scenarios flowchart	flowchart of scenarios in a use case
documentation	additional information about a use case
scenario overview	overview of static information related to a scenario
activities flowchart	flowchart of activities for describing a scenario or another activity (in the case of recursive decomposition)
activity overview	static information related to an activity
role model	overview of role and their relationships
business objects overview	overview of a set of objects used in information exchanges
other	other kind of drawings

6.1.2.20 FurtherActorInformation

Individual or additional information that relates to the use case can be provided.

Table 36 shows all attributes of FurtherActorInformation.

Table 36 – Attributes of UseCaseCoreConcepts::FurtherActorInformation

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 37 shows all association ends of FurtherActorInformation with other classes.

Table 37 – Association ends of UseCaseCoreConcepts::FurtherActorInformation with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] Actor	Actor	information related to an Actor
[0..*]	[0..1] ActorGrouping	ActorGrouping	An Actor is grouped within lists.

6.1.2.21 IdentifiedObject root class

This class is used to gather attributes used for identification purposes. Other classes in this package may inherit from that class.

Table 38 shows all attributes of IdentifiedObject.

Table 38 – Attributes of UseCaseCoreConcepts::IdentifiedObject

name	type	description
description	String	description of the object
mRID	String	master resource identifier
name	String	a short name

6.1.2.22 KeyPerformanceIndicator

Key Performance Indicator (KPI) related to the objectives of the use case

Table 39 shows all attributes of KeyPerformanceIndicator.

Table 39 – Attributes of UseCaseCoreConcepts::KeyPerformanceIndicator

name	type	description
identifier	String	unique identifier for the Key Performance Indicator (KPI)
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 40 shows all association ends of KeyPerformanceIndicator with other classes.

Table 40 – Association ends of UseCaseCoreConcepts::KeyPerformanceIndicator with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1] UseCase	UseCase	the associated use case
[0..1]	[0..*] Objective	Objective	the related objective

6.1.2.23 Narrative

It describes the narrative of the use case.

Table 41 shows all attributes of Narrative.

Table 41 – Attributes of UseCaseCoreConcepts::Narrative

name	type	description
shortDescription	String	It is a short text intended to summarize the main idea as a service for the reader who is searching for a use case or looking for an overview. Recommendation: This short description should have not more than 150 words.
completeDescription	String	It provides a complete narrative of the use case from a user's point of view, describing what occurs when, why, with what expectation, and under what conditions. This narrative should be written in plain text so that non-domain experts can understand it. The length of the complete description can range from a few sentences to a few pages, depending on the complexity and/or newness of the use case. This description often helps the domain expert to reflect on the requirements for the use case before getting into the details in the next sections of the use case template. The description may include drawings for explanation.
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 42 shows all association ends of Narrative with other classes.

Table 42 – Association ends of UseCaseCoreConcepts::Narrative with other classes

[mult from]	[mult to] name	type	description
[0..1]	[1..1] UseCase	UseCase	the associated use case

6.1.2.24 Objective

Objective of the use case

Table 43 shows all attributes of Objective.

Table 43 – Attributes of UseCaseCoreConcepts::Objective

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 44 shows all association ends of Objective with other classes.

Table 44 – Association ends of UseCaseCoreConcepts::Objective with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1] KeyPerformanceIndicator	KeyPerformanceIndicator	the key performance indicator associated with the objective
[0..*]	[0..1] UseCase	UseCase	the associated use case

6.1.2.25 Reference

Any references that might restrict or affect the design, understanding, and requirements of the use case shall be identified, including contracts, regulations, policies, financial considerations, engineering constraints, pollution constraints, and other environmental quality issues.

Table 45 shows all attributes of Reference.

Table 45 – Attributes of UseCaseCoreConcepts::Reference

name	type	description
impact	String	This information describes the nature of the influence of the document on the use case.
link	String	If available, a public link to the document can be provided.
number	String	index of the reference
originatorOrganization	String	This describes the name of the organization publishing the document.
status	String	status of the referenced document
type	String	There are different reference types (e.g. standard, regulation, contract, others like publications).
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 46 shows all association ends of Reference with other classes.

Table 46 – Association ends of UseCaseCoreConcepts::Reference with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] UseCase	UseCase	the associated use case

6.1.2.26 Remark

It is used for further comments which are not considered elsewhere.

Table 47 shows all attributes of Remark.

Table 47 – Attributes of UseCaseCoreConcepts::Remark

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 48 shows all association ends of Remark with other classes.

Table 48 – Association ends of UseCaseCoreConcepts::Remark with other classes

[mult from]	[mult to] name	type	description
[0..*]	[1..1] UseCase	UseCase	the associated use case

6.1.2.27 Requirement

Requirement of a part of the use case

Table 49 shows all attributes of Requirement.

Table 49 – Attributes of UseCaseCoreConcepts::Requirement

name	type	description
identifier	String	identification used in references
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 50 shows all association ends of Requirement with other classes.

Table 50 – Association ends of UseCaseCoreConcepts::Requirement with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] Activity	Activity	the related activity
[0..*]	[0..*] BusinessObject	BusinessObject	the business object associated with the requirement
[0..*]	[0..1] RequirementCategory	RequirementCategory	the associated category of requirements
[0..*]	[0..1] Scenario	Scenario	the related scenario

6.1.2.28 RequirementCategory

Requirements can be sorted in categories. For complex categories the category name can be "dot-delimited".

Table 51 shows all attributes of RequirementCategory.

Table 51 – Attributes of UseCaseCoreConcepts::RequirementCategory

name	type	description
identifier	String	identification used in references
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 52 shows all association ends of RequirementCategory with other classes.

Table 52 – Association ends of UseCaseCoreConcepts::RequirementCategory with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..*] Requirement	Requirement	information about the requirement
[0..1]	[0..*] SubCategory	RequirementCategory	the associated sub categories of requirements
[0..*]	[1..1] RequirementLibrary	RequirementLibrary	the associated library
[0..*]	[0..1] ParentCategory	RequirementCategory	the reference to the parent category of requirement

6.1.2.29 RequirementLibrary

Requirements of the repository

Table 53 shows all attributes of RequirementLibrary.

Table 53 – Attributes of UseCaseCoreConcepts::RequirementLibrary

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 54 shows all association ends of RequirementLibrary with other classes.

Table 54 – Association ends of UseCaseCoreConcepts::RequirementLibrary with other classes

[mult from]	[mult to] name	type	description
[1..1]	[0..*] RequirementCategory	RequirementCategory	the category for the requirement
[0..1]	[1..1] UseCaseRepository	UseCaseRepository	the associated repository

6.1.2.30 Scenario

A possible sequence of interactions

Table 55 shows all attributes of Scenario.

Table 55 – Attributes of UseCaseCoreConcepts::Scenario

name	type	description
number	String	index of the scenario
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 56 shows all association ends of Scenario with other classes.

Table 56 – Association ends of UseCaseCoreConcepts::Scenario with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..*] Activity	Activity	list of activities for the scenario
[0..*]	[0..*] PrimaryActor	Actor	It describes which Actor triggers this scenario.
[0..1]	[0..*] Precondition	Condition	It describes which condition(s) should have been met before this scenario happens.
[0..1]	[0..*] Postcondition	Condition	It describes which condition(s) should prevail after this scenario happens. The post conditions may also define "success" or "failure" conditions for each use case.
[0..1]	[0..*] Drawing	Drawing	the list of associated drawing
[0..1]	[0..*] Requirement	Requirement	information about the requirement
[0..*]	[1..1] UseCase	UseCase	the associated use case
[0..1]	[0..*] TriggeringEvent	TriggeringEvent	list of triggering events for the scenario

6.1.2.31 TriggeringEvent

It describes which event triggers a scenario.

Table 57 shows all attributes of TriggeringEvent.

Table 57 – Attributes of UseCaseCoreConcepts::TriggeringEvent

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 58 shows all association ends of TriggeringEvent with other classes.

Table 58 – Association ends of UseCaseCoreConcepts::TriggeringEvent with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1] Scenario	Scenario	the reference to the scenario having this trigger

6.1.2.32 UseCase

Specification of a set of actions performed by a system, which yields an observable result that is, typically, of value for one or more Actors or other stakeholders of the system

Table 59 shows all attributes of UseCase.

Table 59 – Attributes of UseCaseCoreConcepts::UseCase

name	type	description
classification	String	At the international level the use case description might be generic enough to describe a use case in a more general way independently from the national or regional market design. But use cases might be used to describe regional or national specific circumstances like laws or even project specific details. If the use case reflects those circumstances, it should be characterized accordingly.
identifier	String	The identification number (ID) of a use case is unique within a repository or project and serves for organization/administration of use cases.
keywords	String	Keywords can be defined in order to support extended search functionalities within a use case repository. Multiple keywords should be provided as a comma-separated list.
levelOfDepth	String	Use cases can be described on different levels.
nature	String	It classifies the main focus of the use case.
prioritization	String	If considering a larger number of use cases, it might be interesting to cluster them according to priority. This prioritization might be different from country to country.
scope	String	The scope defines the limits of the use case.
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 60 shows all association ends of UseCase with other classes.

Table 60 – Association ends of UseCaseCoreConcepts::UseCase with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] PrimaryActor	Actor	identification of the PrimaryActor linked to the use case
[0..1]	[0..*] Assumption	Assumption	set of assumptions related to the use case
[0..*]	[0..*] BusinessCase	BusinessCase	list of business cases associated with the use case
[0..*]	[0..*] CommonTerm	CommonTerm	the associated common terms
[1..1]	[0..*] Drawing	Drawing	the list of associated drawings
[1..1]	[0..1] Narrative	Narrative	the information about the narrative part of the use case
[0..1]	[0..*] RelatedObjective	Objective	list of objectives of the use case
[0..*]	[0..*] Reference	Reference	information about referenced documents
[1..1]	[0..*] Remark	Remark	remark associated with the use case
[1..1]	[0..*] Version	Version	list of versions describing changes about the use case
[0..1]	[0..*] RelatedUseCase	UseCase	Known relations to other use cases can be provided here if, for example, the use case is a more detailed one related to a high level use case, or it is an alternative to an existing use case.
[0..*]	[0..*] ActorGrouping	ActorGrouping	A use case references several sets of Actors.
[0..1]	[0..*] Prerequisite	Condition	It describes what condition(s) should have been met before initiation of the use case, such as prior state of the Actors and activities.
[0..1]	[0..*] CustomInformation	CustomInformation	related additional information
[0..1]	[0..*] KeyPerformanceIndicator	KeyPerformanceIndicator	information about the KPI related to the use case
[1..1]	[0..*] Scenario	Scenario	reference to the scenarios belonging to the use case
[0..*]	[1..1] UseCaseLibrary	UseCaseLibrary	reference to the associated library
[0..*]	[0..1] UseCase	UseCase	parent use case

6.1.2.33 UseCaseLibrary

Use cases of the repository

Table 61 shows all attributes of UseCaseLibrary.

Table 61 – Attributes of UseCaseCoreConcepts::UseCaseLibrary

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 62 shows all association ends of UseCaseLibrary with other classes.

Table 62 – Association ends of UseCaseCoreConcepts::UseCaseLibrary with other classes

[mult from]	[mult to] name	type	description
[1..1]	[0..*] UseCase	UseCase	list of use cases within the library
[0..1]	[1..1] UseCaseRepository	UseCaseRepository	the associated repository

6.1.2.34 UseCaseRepository

Database, based on a given use cases template, for editing, maintenance and administration of use cases, Actors and requirements including their interrelations

Table 63 shows all attributes of UseCaseRepository.

Table 63 – Attributes of UseCaseCoreConcepts::UseCaseRepository

name	type	description
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 64 shows all association ends of UseCaseRepository with other classes.

Table 64 – Association ends of UseCaseCoreConcepts::UseCaseRepository with other classes

[mult from]	[mult to] name	type	description
[1..1]	[0..*] ActorLibrary	ActorLibrary	library of Actors
[1..1]	[0..1] AreaLibrary	AreaLibrary	the associated library of areas
[1..1]	[0..1] BusinessObjectLibrary	BusinessObjectLibrary	the library of business objects
[1..1]	[0..1] RequirementLibrary	RequirementLibrary	the associated library
[1..1]	[0..1] UseCaseLibrary	UseCaseLibrary	the associated library of use cases
[0..1]	[0..1] BusinessCaseLibrary	BusinessCaseLibrary	the library associated with the repository
[1..1]	[0..*] CommonTermLibrary	CommonTermLibrary	the associated library of common terms

6.1.2.35 Version

Version of the use case description

Table 65 shows all attributes of Version.

Table 65 – Attributes of UseCaseCoreConcepts::Version

name	type	description
approvalStatus	String	information used within the standardization process
changes	String	It represents the differences with respect to the previous version. Multiple changes are separated with paragraphs.
date	DateTime	date of creation of the version, in the format YYYY-MM-DD
number	String	sequential number to identify the version of the document
description	String	inherited from: IdentifiedObject
mRID	String	inherited from: IdentifiedObject
name	String	inherited from: IdentifiedObject

Table 66 shows all association ends of Version with other classes.

Table 66 – Association ends of UseCaseCoreConcepts::Version with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..*] Author	Author	list of authors participating in the changes of this version
[0..*]	[1..1] UseCase	UseCase	information about the associated use case

6.1.2.36 Top package Primitives

6.1.2.36.1 General

Figure 17 shows class diagram Primitives.

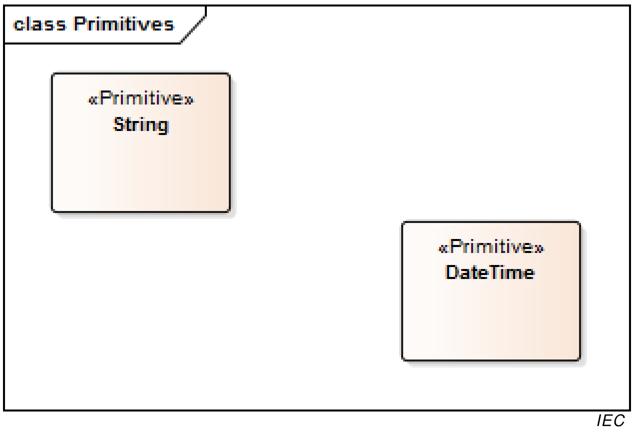

Figure 17 – Class diagram Primitives::Primitives

6.1.2.36.2 DateTime primitive

DateTime Primitive type

6.1.2.36.3 String primitive

String Primitive type

6.2 Use case library exchange format

6.2.1 UML use case library exchange model (UML Contextual Model)

6.2.1.1 General

The use case library exchange model is described through a UML Contextual Model. The XSD content is generated from this model.

6.2.1.2 Model package UseCase_ExchangeModel

6.2.1.2.1 General

IEC 62559 UML Contextual Model for the use case information exchange model

Figure 18 shows package diagram ContextualModel_Dependency.

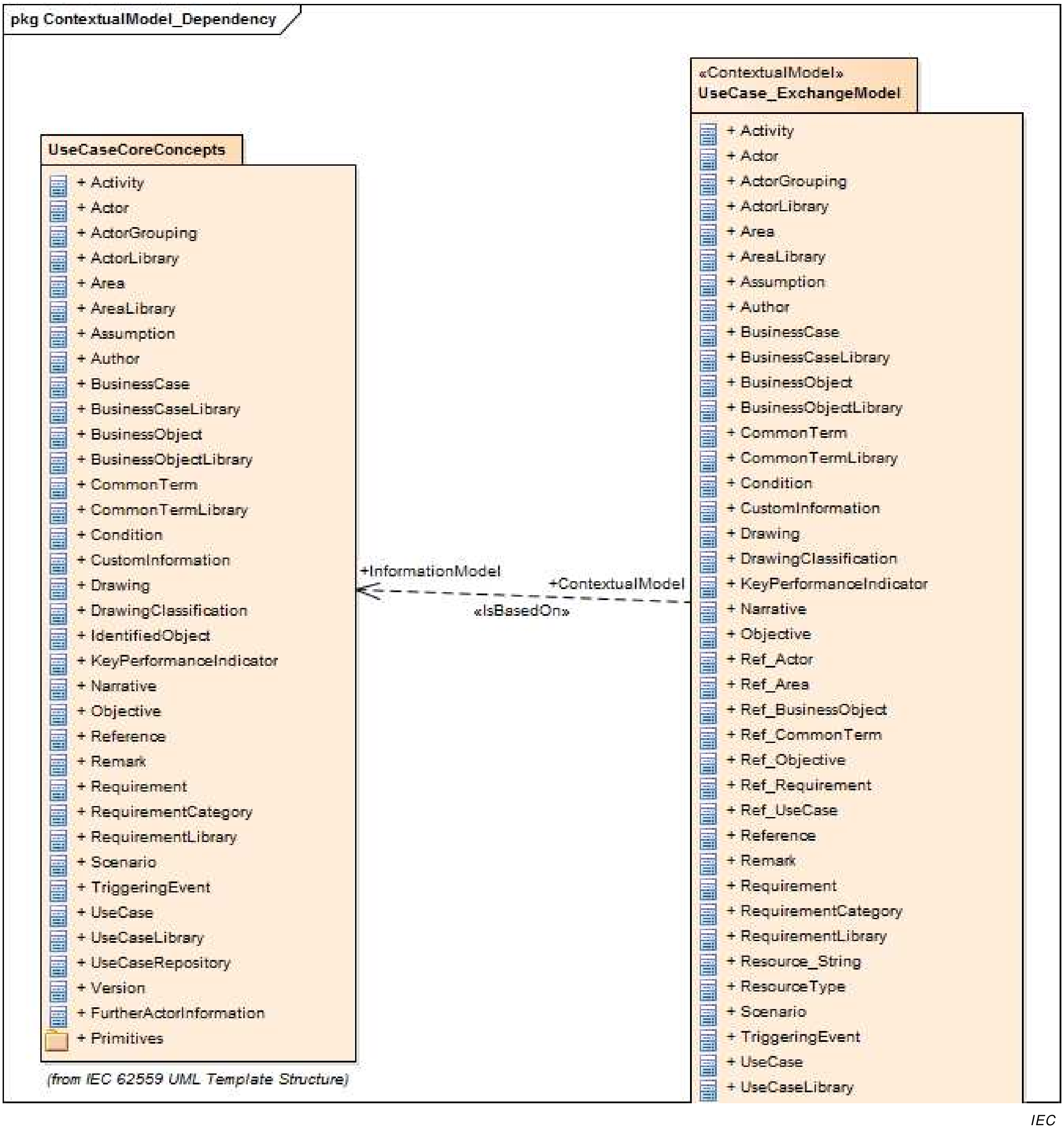

Figure 18 – Package diagram UseCase_ExchangeModel::ContextualModel_Dependency

Figure 19 shows class diagram UseCase_Repository.

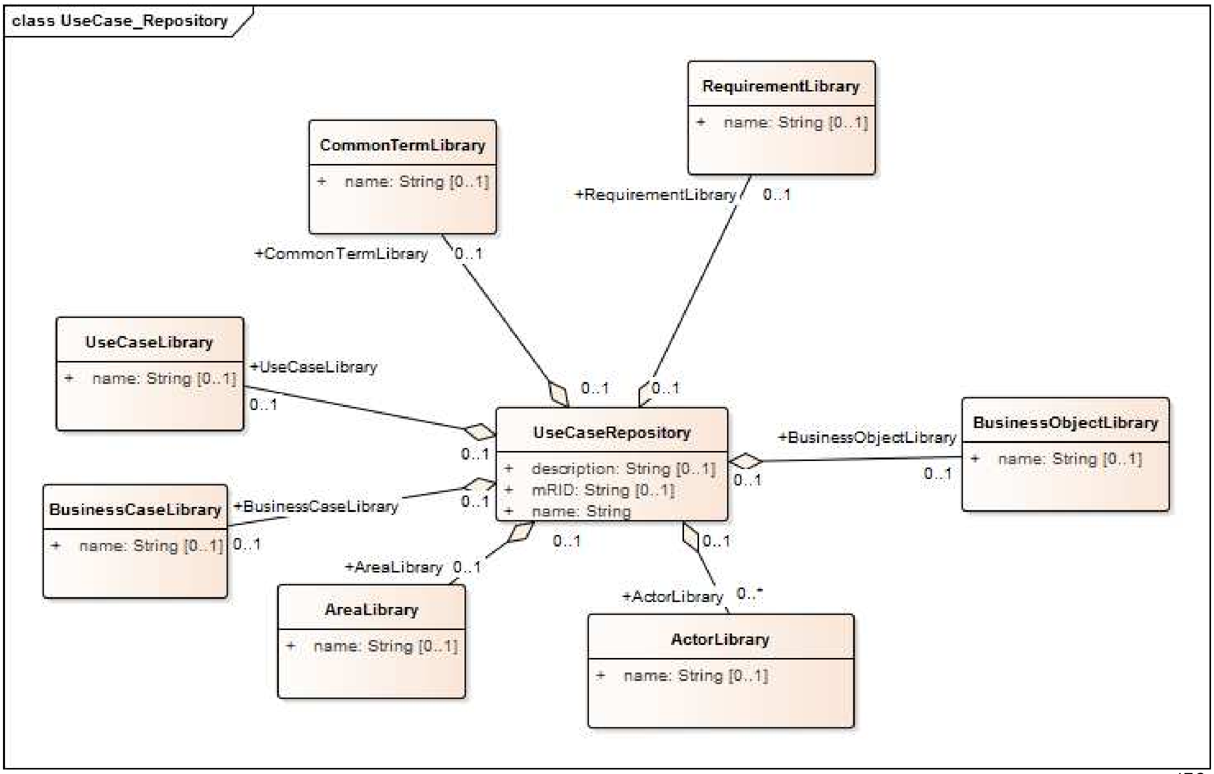

Figure 19 – Class diagram UseCase_ExchangeModel::UseCase_Repository

Figure 20 shows class diagram UseCase_ShortDescription.

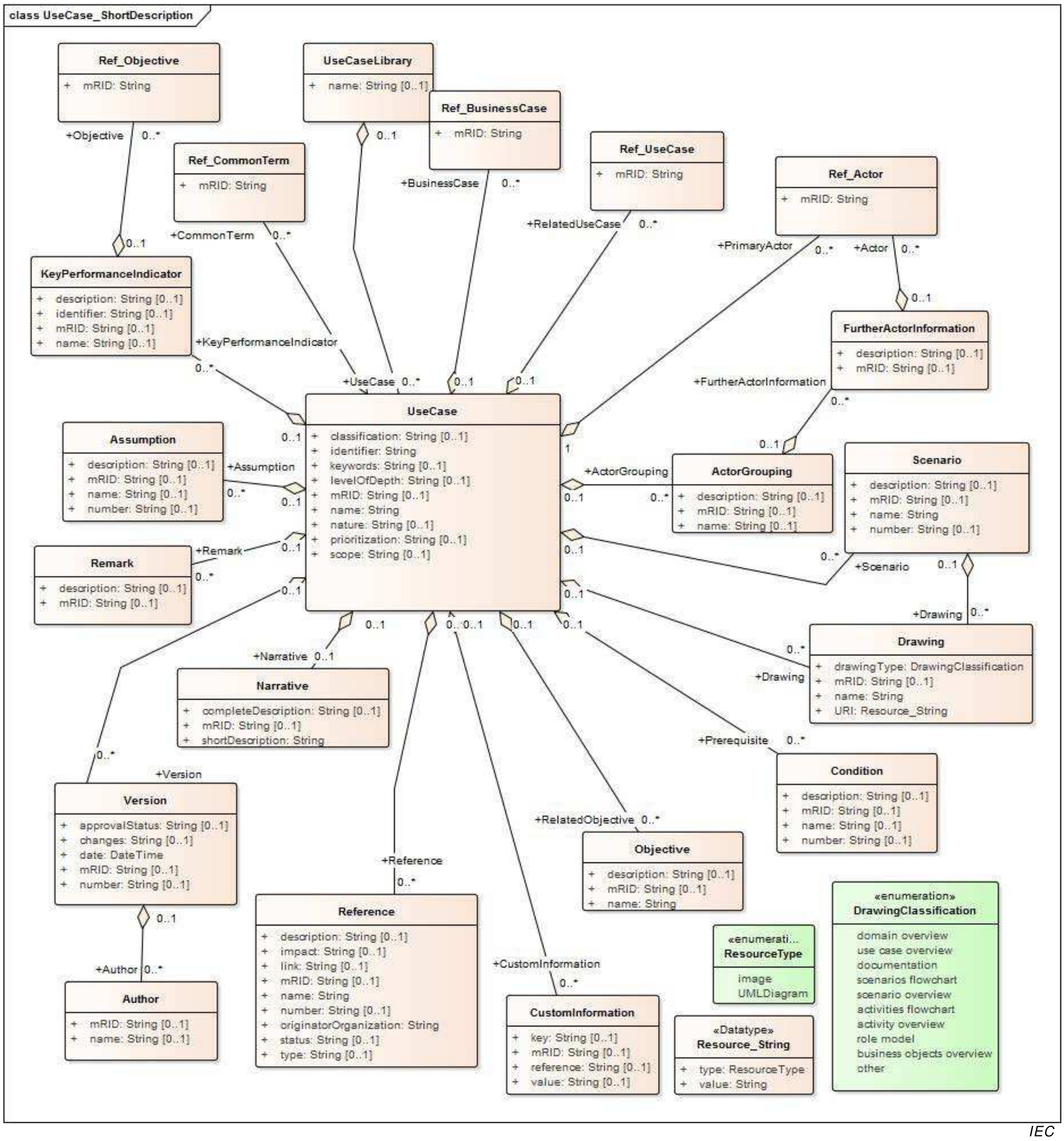

Figure 20 – Class diagram UseCase_ExchangeModel::UseCase_ShortDescription

Figure 21 shows class diagram UseCase_DetailedActivities.

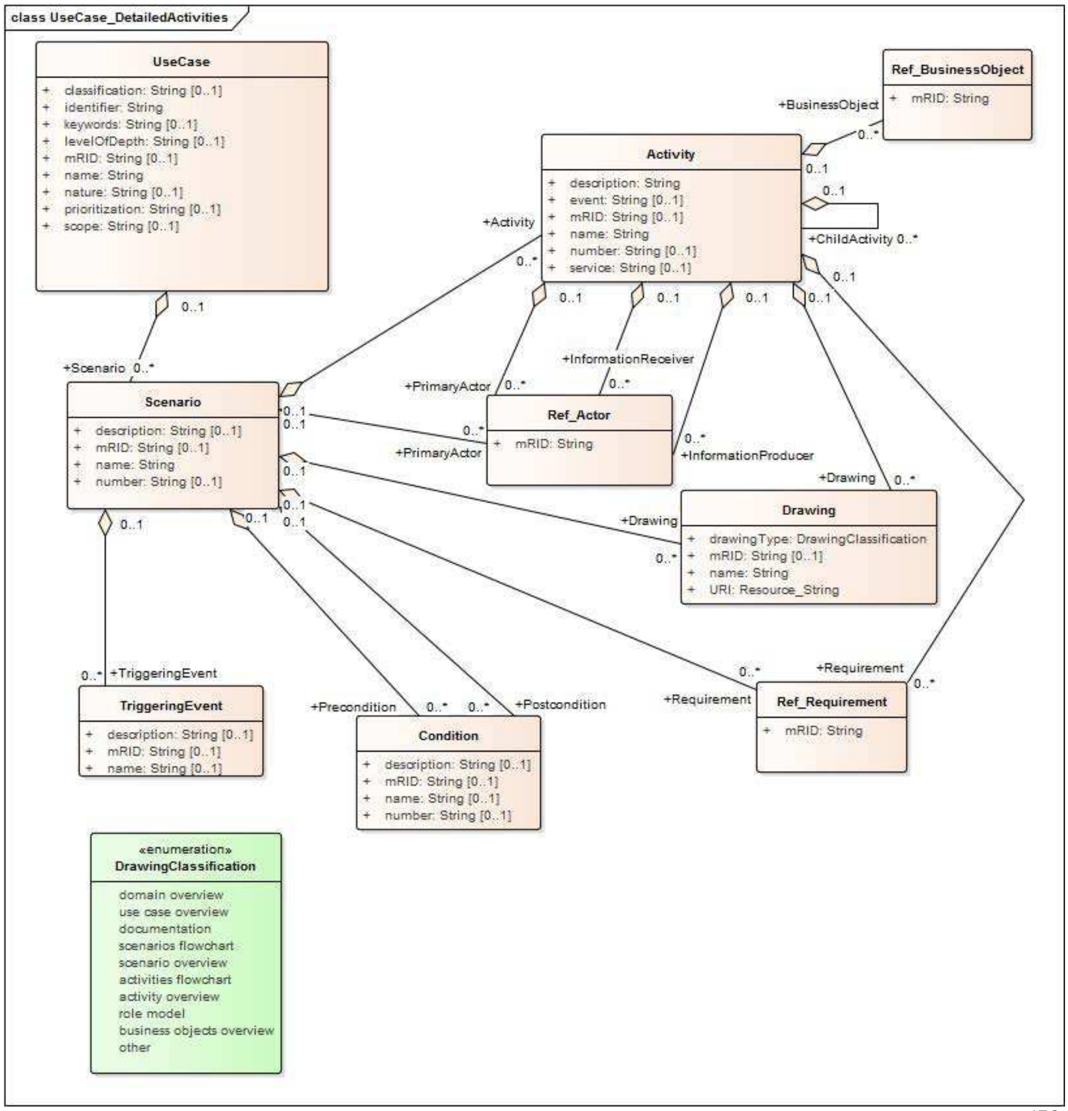

Figure 21 – Class diagram UseCase_ExchangeModel::UseCase_DetailedActivities

Figure 22 shows class diagram UseCase_CompleteDescription.

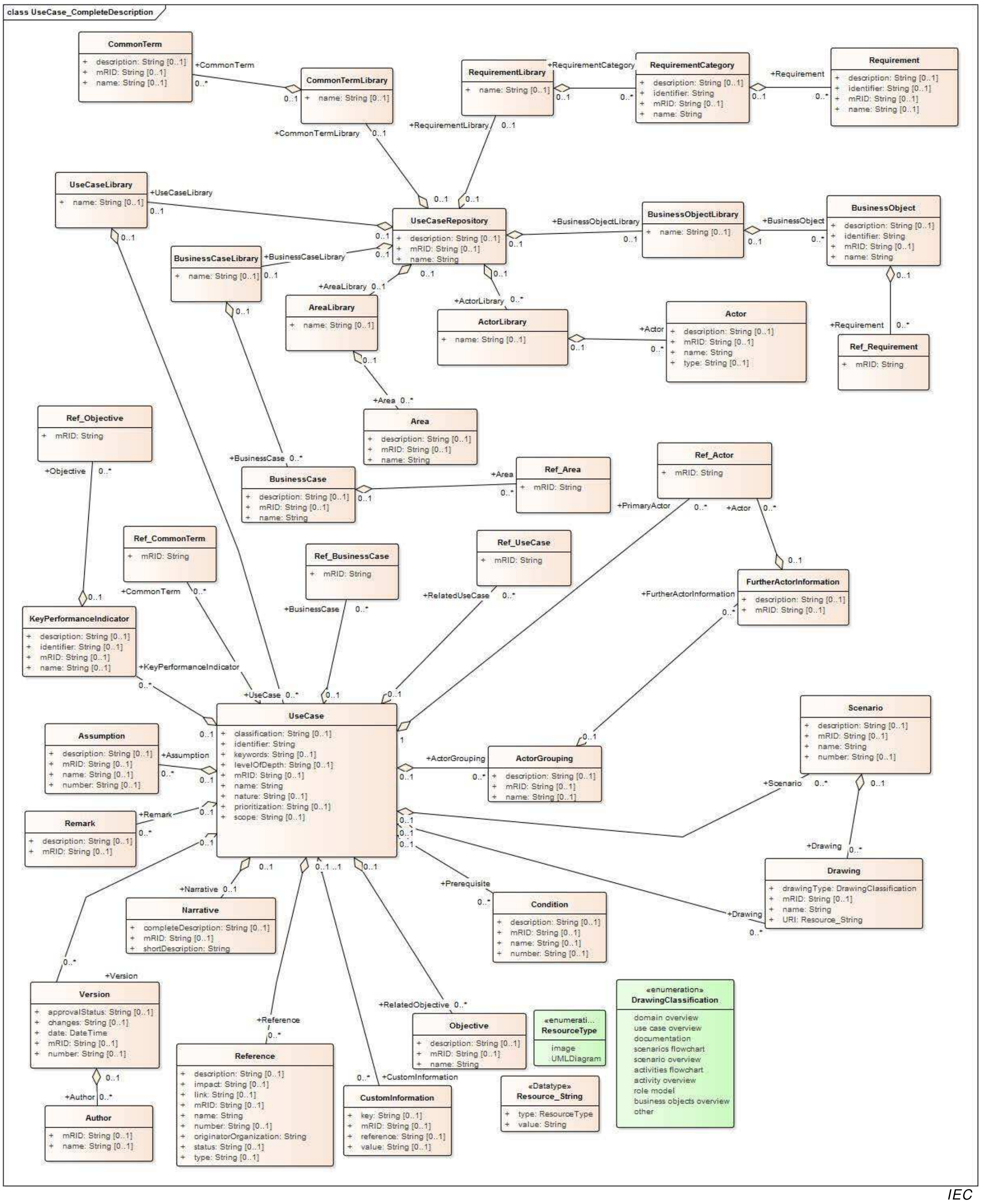

Figure 22 – Class diagram UseCase_ExchangeModel::UseCase_CompleteDescription

6.2.1.2.2 Activity root class

It may be subdivided into other activities or represent an elementary step of the scenario.

Table 67 shows all attributes of Activity.

Table 67 – Attributes of UseCase_ExchangeModel::Activity

name	type	description
name	String	a short name
description	String	description of the object
event	String	It is the event that triggers the step. This might be completion of previous steps (in that case the event is the comma-separated list of their numbers) or the condition to exit a loop.
service	String	This column identifies the nature of the information flow and the originator of the information. Available options are CREATE, GET, CHANGE, DELETE, CANCEL, EXECUTE derived from IEC 61968-100:2013. Additionally, REPORT, TIMER and REPEAT are suggested. – CREATE means that an information object is to be created at the Producer. – GET (this is the default value if none is populated) means that the Receiver requests information from the Producer (default). – CHANGE means that information is to be updated. Producer updates the Receiver's information. – DELETE means that information is to be deleted. Producer deletes information from the Receiver. – CANCEL, CLOSE imply actions related to processes, such as the closure of a work order or the cancellation of a control request. – EXECUTE is used when a complex transaction is being conveyed using a service, which potentially contains more than one verb. – REPORT is used to represent transferral of unsolicited information or asynchronous information flows. Producer provides information to the Receiver. – TIMER is used to represent a waiting period. When using the TIMER service, the Information Producer and Information Receiver fields shall refer to the same Actor. – REPEAT is used to indicate that a series of steps is repeated until a condition or trigger event. The condition is specified as the text in the "Event" column for this row or step. Following the word REPEAT, the first and last step numbers of the series to be repeated shall appear, in parenthesis, in the following form: REPEAT(X-Y) where X is the first step and Y is the last step. These common service definitions are related to automation/information or communication systems. In case the use case template is applied in other domains, further services might be used and described.
mRID	String	master resource identifier

Table 68 shows all association ends of Activity with other classes.

Table 68 – Association ends of UseCase_ExchangeModel::Activity with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	Scenario	This association end is an aggregation.
[0..*]	[0..1]	Activity	This association end is an aggregation.
[0..1]	[0..*] InformationReceiver	Ref_Actor	This identifies the receiver of the information. This should be one of the Actors defined in the repository.
[0..1]	[0..*] InformationProducer	Ref_Actor	This identifies the producer or source of the information. This should be one of the Actors defined in the repository.
[0..1]	[0..*] Requirement	Ref_Requirement	Several requirements may be listed in one step, comma separated.
[0..1]	[0..*] BusinessObject	Ref_BusinessObject	This describes briefly the information to be exchanged between the two Actors – information producer and information receiver. It may list several information items in one step, comma separated.
[0..1]	[0..*] Drawing	Drawing	the list of associated drawing
[0..1]	[0..*] PrimaryActor	Ref_Actor	reference to the PrimaryActor
[0..1]	[0..*] ChildActivity	Activity	child activities

6.2.1.2.3 Actor root class

It is the entity that communicates and interacts.

These Actors can include people, software applications, systems, databases, and even the power system itself.

Table 69 shows all attributes of Actor.

Table 69 – Attributes of UseCase_ExchangeModel::Actor

name	type	description
name	String	a short name
type	String	type of the Actor (business, system, ...)
mRID	String	master resource identifier

Table 70 shows all association ends of Actor with other classes.

Table 70 – Association ends of UseCase_ExchangeModel::Actor with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	ActorLibrary	This association end is an aggregation.

6.2.1.2.4 ActorGrouping root class

Group of Actors used to organize an Actor list

Table 71 shows all attributes of ActorGrouping.

Table 71 – Attributes of UseCase_ExchangeModel::ActorGrouping

name	type	description
name	String	a short name
description	String	description of the object
mRID	String	master resource identifier

Table 72 shows all association ends of ActorGrouping with other classes.

Table 72 – Association ends of UseCase_ExchangeModel::ActorGrouping with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..*] FurtherActorInformation	FurtherActorInformation	In a context of one use case, an Actor grouped into an ActorGrouping may have one FurtherActorInformation.
[0..*]	[0..1]	UseCase	This association end is an aggregation.

6.2.1.2.5 ActorLibrary root class

Actors of the repository

Table 73 shows all attributes of ActorLibrary.

Table 73 – Attributes of UseCase_ExchangeModel::ActorLibrary

name	type	description
name	String	a short name

Table 74 shows all association ends of ActorLibrary with other classes.

Table 74 – Association ends of UseCase_ExchangeModel::ActorLibrary with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	UseCaseRepository	This association end is an aggregation.
[0..1]	[0..*] Actor	Actor	the Actors listed in the library

6.2.1.2.6 Area root class

Area of knowledge or activity characterized by a set of concepts and terminology understood by the practitioners in that area

Table 75 shows all attributes of Area.

Table 75 – Attributes of UseCase_ExchangeModel::Area

name	type	description
name	String	a short name
description	String	description of the object
mRID	String	master resource identifier

Table 76 shows all association ends of Area with other classes.

Table 76 – Association ends of UseCase_ExchangeModel::Area with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	AreaLibrary	This association end is an aggregation.

6.2.1.2.7 AreaLibrary root class

Areas of the repository

Table 77 shows all attributes of AreaLibrary.

Table 77 – Attributes of UseCase_ExchangeModel::AreaLibrary

name	type	description
name	String	a short name

Table 78 shows all association ends of AreaLibrary with other classes.

Table 78 – Association ends of UseCase_ExchangeModel::AreaLibrary with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..1]	UseCaseRepository	This association end is an aggregation.
[0..1]	[0..*] Area	Area	the list of areas within the library

6.2.1.2.8 Assumption root class

It may be used to define a further general assumption for a use case, such as which systems already exist, which contractual relations exist, and which configurations of systems are probably in place. Initial states of information exchanged shall also be identified.

Table 79 shows all attributes of Assumption.

Table 79 – Attributes of UseCase_ExchangeModel::Assumption

name	type	description
name	String	a short name
number	String	index of the assumption
description	String	description of the object
mRID	String	master resource identifier

Table 80 shows all association ends of Assumption with other classes.

Table 80 – Association ends of UseCase_ExchangeModel::Assumption with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	UseCase	This association end is an aggregation.

6.2.1.2.9 Author root class

This field is used to document who has provided the current version. It can be a person, organization or, for example, standardization committee like a technical committee (TC) or working group (WG).

Table 81 shows all attributes of Author.

Table 81 – Attributes of UseCase_ExchangeModel::Author

name	type	description
name	String	a short name
mRID	String	master resource identifier

Table 82 shows all association ends of Author with other classes.

Table 82 – Association ends of UseCase_ExchangeModel::Author with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	Version	This association end is an aggregation.

6.2.1.2.10 BusinessCase root class

It provides a description or reference with some rationale for the suggested use case. Usually the business case is related to several use cases.

Table 83 shows all attributes of BusinessCase.

Table 83 – Attributes of UseCase_ExchangeModel::BusinessCase

name	type	description
name	String	a short name
description	String	description of the object
mRID	String	master resource identifier

Table 84 shows all association ends of BusinessCase with other classes.

Table 84 – Association ends of UseCase_ExchangeModel::BusinessCase with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	BusinessCaseLibrary	This association end is an aggregation.
[0..1]	[0..*] Area	Ref_Area	Use cases can be used in various areas (e.g. energy system). Within these areas different domains are used to define / determine a more specific subgrouping. One or more domains and zones, comma separated, can be specified in the template.

6.2.1.2.11 BusinessCaseLibrary root class

Business cases of the repository

Table 85 shows all attributes of BusinessCaseLibrary.

Table 85 – Attributes of UseCase_ExchangeModel::BusinessCaseLibrary

name	type	description
name	String	a short name

Table 86 shows all association ends of BusinessCaseLibrary with other classes.

Table 86 – Association ends of UseCase_ExchangeModel::BusinessCaseLibrary with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..1]	UseCaseRepository	This association end is an aggregation.
[0..1]	[0..*] BusinessCase	BusinessCase	the list of business cases

6.2.1.2.12 BusinessObject root class

Business object exchanged in detailed activity steps

Table 87 shows all attributes of BusinessObject.

Table 87 – Attributes of UseCase_ExchangeModel::BusinessObject

name	type	description
name	String	a short name
description	String	description of the object
identifier	String	identification used in references in the use case description
mRID	String	master resource identifier

Table 88 shows all association ends of BusinessObject with other classes.

Table 88 – Association ends of UseCase_ExchangeModel::BusinessObject with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	BusinessObjectLibrary	This association end is an aggregation.
[0..1]	[0..*] Requirement	Ref_Requirement	It can be used to define requirements related to the information and not to the step as in the step by step analysis.

6.2.1.2.13 BusinessObjectLibrary root class

Business objects which are exchanged in use cases

Table 89 shows all attributes of BusinessObjectLibrary.

Table 89 – Attributes of UseCase_ExchangeModel::BusinessObjectLibrary

name	type	description
name	String	a short name

Table 90 shows all association ends of BusinessObjectLibrary with other classes.

Table 90 – Association ends of UseCase_ExchangeModel::BusinessObjectLibrary with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..1]	UseCaseRepository	This association end is an aggregation.
[0..1]	[0..*] BusinessObject	BusinessObject	list of business objects within the library

6.2.1.2.14 CommonTerm root class

It represents a glossary term and its definition.

Table 91 shows all attributes of CommonTerm.

Table 91 – Attributes of UseCase_ExchangeModel::CommonTerm

name	type	description
name	String	a short name
description	String	description of the object
mRID	String	master resource identifier

Table 92 shows all association ends of CommonTerm with other classes.

Table 92 – Association ends of UseCase_ExchangeModel::CommonTerm with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	CommonTermLibrary	This association end is an aggregation.

6.2.1.2.15 CommonTermLibrary root class

Common glossary defining terms for all use cases.

Table 93 shows all attributes of CommonTermLibrary.

Table 93 – Attributes of UseCase_ExchangeModel::CommonTermLibrary

name	type	description
name	String	a short name

Table 94 shows all association ends of CommonTermLibrary with other classes.

Table 94 – Association ends of UseCase_ExchangeModel::CommonTermLibrary with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..1]	UseCaseRepository	This association end is an aggregation.
[0..1]	[0..*] CommonTerm	CommonTerm	list of common terms

6.2.1.2.16 Condition root class

It describes either a precondition or a postcondition.

Table 95 shows all attributes of Condition.

Table 95 – Attributes of UseCase_ExchangeModel::Condition

name	type	description
name	String	a short name
number	String	index of the condition
description	String	description of the object
mRID	String	master resource identifier

Table 96 shows all association ends of Condition with other classes.

Table 96 – Association ends of UseCase_ExchangeModel::Condition with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	Scenario	This association end is an aggregation.
[0..*]	[0..1]	Scenario	This association end is an aggregation.
[0..*]	[0..1]	UseCase	This association end is an aggregation.

6.2.1.2.17 CustomInformation root class

It provides a flexible option to include miscellaneous custom, semi-structured information which does not fit into other parts of the template.

Table 97 shows all attributes of CustomInformation.

Table 97 – Attributes of UseCase_ExchangeModel::CustomInformation

name	type	description
key	String	unique key for identification purposes
value	String	the corresponding information for the provided key
reference	String	This field refers to relevant template section.
mRID	String	master resource identifier

Table 98 shows all association ends of CustomInformation with other classes.

Table 98 – Association ends of UseCase_ExchangeModel::CustomInformation with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	UseCase	This association end is an aggregation.

6.2.1.2.18 Drawing root class

For clarification, in general it is recommended to provide drawing(s) by hand, by a graphic or as UML graphics (preferred in IEC 62559). The drawing should show interactions which identify the steps where possible.

Table 99 shows all attributes of Drawing.

Table 99 – Attributes of UseCase_ExchangeModel::Drawing

name	type	description
name	String	a short name
drawingType	DrawingClassification	type of drawing
URI	Resource_String	resource Path to an image file ex: http://www…./image.jpg or path to an XMI file with a relative diagram path from the root of the UML model. ex: http://www.../Model.xmi#Package1/Package2/.../DiagName
mRID	String	master resource identifier

Table 100 shows all association ends of Drawing with other classes.

Table 100 – Association ends of UseCase_ExchangeModel::Drawing with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	Activity	This association end is an aggregation.
[0..*]	[0..1]	UseCase	This association end is an aggregation.
[0..*]	[0..1]	Scenario	This association end is an aggregation.

6.2.1.2.19 DrawingClassification enumeration

Possible types of drawing.

Table 101 shows all literals of DrawingClassification.

Table 101 – Literals of UseCase_ExchangeModel::DrawingClassification

literal	description
domain overview	overview of relationships between domains and use cases
use case overview	overview of use case short template information
documentation	additional information about a use case
scenarios flowchart	flowchart of scenarios in a use case
scenario overview	overview of static information related to a scenario
activities flowchart	flowchart of activities for describing a scenario or another activity (in the case of recursive decomposition)
activity overview	static information related to an activity
role model	overview of role and their relationships
business objects overview	overview of a set of objects used in information exchanges
other	other kind of drawings

6.2.1.2.20 KeyPerformanceIndicator root class

Key Performance Indicator (KPI) related to the objectives of the use case.

Table 102 shows all attributes of KeyPerformanceIndicator.

Table 102 – Attributes of UseCase_ExchangeModel::KeyPerformanceIndicator

name	type	description
name	String	a short name
description	String	description of the object
identifier	String	unique identifier for the Key Performance Indicator (KPI)
mRID	String	master resource identifier

Table 103 shows all association ends of KeyPerformanceIndicator with other classes.

Table 103 – Association ends of UseCase_ExchangeModel::KeyPerformanceIndicator with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	UseCase	This association end is an aggregation.
[0..1]	[0..*] Objective	Ref_Objective	the related objective

6.2.1.2.21 Narrative root class

It describes the narrative of the use case.

Table 104 shows all attributes of Narrative.

Table 104 – Attributes of UseCase_ExchangeModel::Narrative

name	type	description
shortDescription	String	It is a short text intended to summarize the main idea as a service for the reader who is searching for a use case or looking for an overview. Recommendation: This short description should have not more than 150 words.
completeDescription	String	It provides a complete narrative of the use case from a user's point of view, describing what occurs when, why, with what expectation, and under what conditions. This narrative should be written in plain text so that non-domain experts can understand it. The length of the complete description can range from a few sentences to a few pages, depending on the complexity and/or newness of the use case. This description often helps the domain expert to reflect on the requirements for the use case before getting into the details in the next sections of the use case template. The description may include drawings for explanation.
mRID	String	master resource identifier

Table 105 shows all association ends of Narrative with other classes.

Table 105 – Association ends of UseCase_ExchangeModel::Narrative with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..1]	UseCase	This association end is an aggregation.

6.2.1.2.22 Objective root class

Objective of the use case.

Table 106 shows all attributes of Objective.

Table 106 – Attributes of UseCase_ExchangeModel::Objective

name	type	description
name	String	a short name
description	String	description of the object
mRID	String	master resource identifier

Table 107 shows all association ends of Objective with other classes.

Table 107 – Association ends of UseCase_ExchangeModel::Objective with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	UseCase	This association end is an aggregation.

6.2.1.2.23 Ref_Actor root class

It is the entity that communicates and interacts.

These Actors can include people, software applications, systems, databases, and even the power system itself.

Table 108 shows all attributes of Ref_Actor.

Table 108 – Attributes of UseCase_ExchangeModel::Ref_Actor

name	type	description
mRID	String	master resource identifier

Table 109 shows all association ends of Ref_Actor with other classes.

Table 109 – Association ends of UseCase_ExchangeModel::Ref_Actor with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	Activity	This association end is an aggregation.
[0..*]	[0..1]	Activity	This association end is an aggregation.
[0..*]	[0..1]	Activity	This association end is an aggregation.
[0..*]	[0..1]	Scenario	This association end is an aggregation.
[0..*]	[0..1]	FurtherActorInformation	This association end is an aggregation.
[0..*]	[1..1]	UseCase	This association end is an aggregation.

6.2.1.2.24 Ref_Area root class

Area of knowledge or activity characterized by a set of concepts and terminology understood by the practitioners in that area.

Table 110 shows all attributes of Ref_Area.

Table 110 – Attributes of UseCase_ExchangeModel::Ref_Area

name	type	description
mRID	String	master resource identifier

Table 111 shows all association ends of Ref_Area with other classes.

Table 111 – Association ends of UseCase_ExchangeModel::Ref_Area with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	BusinessCase	This association end is an aggregation.

6.2.1.2.25 Ref_BusinessCase root class

It provides a description or reference with some rationale for the suggested use case. Usually the business case is related to several use cases.

Table 112 shows all attributes of Ref_BusinessCase.

Table 112 – Attributes of UseCase_ExchangeModel::Ref_BusinessCase

name	type	description
mRID	String	master resource identifier

Table 113 shows all association ends of Ref_BusinessCase with other classes.

Table 113 – Association ends of UseCase_ExchangeModel::Ref_BusinessCase with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	UseCase	This association end is an aggregation.

6.2.1.2.26 Ref_BusinessObject root class

Business object exchanged in detailed activity steps.

Table 114 shows all attributes of Ref_BusinessObject.

Table 114 – Attributes of UseCase_ExchangeModel::Ref_BusinessObject

name	type	description
mRID	String	master resource identifier

Table 115 shows all association ends of Ref_BusinessObject with other classes.

Table 115 – Association ends of UseCase_ExchangeModel::Ref_BusinessObject with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	Activity	This association end is an aggregation.

6.2.1.2.27 Ref_CommonTerm root class

It represents a glossary term and its definition.

Table 116 shows all attributes of Ref_CommonTerm.

Table 116 – Attributes of UseCase_ExchangeModel::Ref_CommonTerm

name	type	description
mRID	String	master resource identifier

Table 117 shows all association ends of Ref_CommonTerm with other classes.

Table 117 – Association ends of UseCase_ExchangeModel::Ref_CommonTerm with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	UseCase	This association end is an aggregation.

6.2.1.2.28 Ref_Objective root class

Objective of the use case.

Table 118 shows all attributes of Ref_Objective.

Table 118 – Attributes of UseCase_ExchangeModel::Ref_Objective

name	type	description
mRID	String	master resource identifier

Table 119 shows all association ends of Ref_Objective with other classes.

Table 119 – Association ends of UseCase_ExchangeModel::Ref_Objective with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	KeyPerformanceIndicator	This association end is an aggregation.

6.2.1.2.29 Ref_Requirement root class

Requirement of a part of the use case.

Table 120 shows all attributes of Ref_Requirement.

Table 120 – Attributes of UseCase_ExchangeModel::Ref_Requirement

name	type	description
mRID	String	master resource identifier

Table 121 shows all association ends of Ref_Requirement with other classes.

Table 121 – Association ends of UseCase_ExchangeModel::Ref_Requirement with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	Activity	This association end is an aggregation.
[0..*]	[0..1]	BusinessObject	This association end is an aggregation.
[0..*]	[0..1]	Scenario	This association end is an aggregation.

6.2.1.2.30 Ref_UseCase root class

Specification of a set of actions performed by a system, which yields an observable result that, is, typically, of value for one or more Actors or other stakeholders of the system.

Table 122 shows all attributes of Ref_UseCase.

Table 122 – Attributes of UseCase_ExchangeModel::Ref_UseCase

name	type	description
mRID	String	master resource identifier

Table 123 shows all association ends of Ref_UseCase with other classes.

Table 123 – Association ends of UseCase_ExchangeModel::Ref_UseCase with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	UseCase	This association end is an aggregation.

6.2.1.2.31 Reference root class

Any references that might restrict or affect the design, understanding, and requirements of the use case shall be identified, including contracts, regulations, policies, financial considerations, engineering constraints, pollution constraints, and other environmental quality issues.

Table 124 shows all attributes of Reference.

Table 124 – Attributes of UseCase_ExchangeModel::Reference

name	type	description
name	String	a short name
description	String	description of the object
type	String	There are different reference types (e.g. standard, regulation, contract, others like publications).
impact	String	This information describes the nature of the influence of the document on the use case.
status	String	status of the referenced document
link	String	If available, a public link to the document can be provided.
originatorOrganization	String	This describes the name of the organization publishing the document.
number	String	index of the reference
mRID	String	master resource identifier

Table 125 shows all association ends of Reference with other classes.

Table 125 – Association ends of UseCase_ExchangeModel::Reference with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	UseCase	This association end is an aggregation.

6.2.1.2.32 Remark root class

It is used for further comments which are not considered elsewhere.

Table 126 shows all attributes of Remark.

Table 126 – Attributes of UseCase_ExchangeModel::Remark

name	type	description
description	String	description of the object
mRID	String	master resource identifier

Table 127 shows all association ends of Remark with other classes.

Table 127 – Association ends of UseCase_ExchangeModel::Remark with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	UseCase	This association end is an aggregation.

6.2.1.2.33 Requirement root class

Requirement of a part of the use case.

Table 128 shows all attributes of Requirement.

Table 128 – Attributes of UseCase_ExchangeModel::Requirement

name	type	description
name	String	a short name
identifier	String	identification used in references
description	String	description of the object
mRID	String	master resource identifier

Table 129 shows all association ends of Requirement with other classes.

Table 129 – Association ends of UseCase_ExchangeModel::Requirement with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	RequirementCategory	This association end is an aggregation.

6.2.1.2.34 RequirementCategory root class

Requirements can be sorted in categories. For complex categories the category name can be "dot-delimited".

Table 130 shows all attributes of RequirementCategory.

Table 130 – Attributes of UseCase_ExchangeModel::RequirementCategory

name	type	description
name	String	a short name
identifier	String	identification used in references
description	String	description of the object
mRID	String	master resource identifier

Table 131 shows all association ends of RequirementCategory with other classes.

Table 131 – Association ends of UseCase_ExchangeModel::RequirementCategory with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	RequirementLibrary	This association end is an aggregation.
[0..1]	[0..*] Requirement	Requirement	information about the requirement

6.2.1.2.35 RequirementLibrary root class

Requirements of the repository.

Table 132 shows all attributes of RequirementLibrary.

Table 132 – Attributes of UseCase_ExchangeModel::RequirementLibrary

name	type	description
name	String	a short name

Table 133 shows all association ends of RequirementLibrary with other classes.

Table 133 – Association ends of UseCase_ExchangeModel::RequirementLibrary with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..1]	UseCaseRepository	This association end is an aggregation.
[0..1]	[0..*] RequirementCategory	RequirementCategory	the category for the requirement

6.2.1.2.36 Resource_String datatype

String Primitive type.

Table 134 shows all attributes of Resource_String.

Table 134 – Attributes of UseCase_ExchangeModel::Resource_String

name	type	description
value	String	primary value of the data type
type	ResourceType	supplementary attribute of the data type

6.2.1.2.37 ResourceType enumeration

List of possible resources.

Table 135 shows all literals of ResourceType.

Table 135 – Literals of UseCase_ExchangeModel::ResourceType

literal	description
UMLDiagram	UML kind for a resource

6.2.1.2.38 Scenario root class

A possible sequence of interactions.

Table 136 shows all attributes of Scenario.

Table 136 – Attributes of UseCase_ExchangeModel::Scenario

name	type	description
name	String	a short name
description	String	description of the object
number	String	index of the scenario
mRID	String	master resource identifier

Table 137 shows all association ends of Scenario with other classes.

Table 137 – Association ends of UseCase_ExchangeModel::Scenario with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..*] Requirement	Ref_Requirement	information about the requirement
[0..*]	[0..1]	UseCase	This association end is an aggregation.
[0..1]	[0..*] Activity	Activity	list of activities for the scenario
[0..1]	[0..*] Precondition	Condition	It describes which condition(s) should have been met before this scenario happens.
[0..1]	[0..*] Postcondition	Condition	It describes which condition(s) should prevail after this scenario happens. The post conditions may also define "success" or "failure" conditions for each use case.
[0..1]	[0..*] Drawing	Drawing	the list of associated drawing
[0..1]	[0..*] PrimaryActor	Ref_Actor	It describes which Actor triggers this scenario.
[0..1]	[0..*] TriggeringEvent	TriggeringEvent	list of triggering events for the scenario

6.2.1.2.39 TriggeringEvent root class

It describes which event triggers a scenario.

Table 138 shows all attributes of TriggeringEvent.

Table 138 – Attributes of UseCase_ExchangeModel::TriggeringEvent

name	type	description
name	String	a short name
description	String	description of the object
mRID	String	master resource identifier

Table 139 shows all association ends of TriggeringEvent with other classes.

Table 139 – Association ends of UseCase_ExchangeModel::TriggeringEvent with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	Scenario	This association end is an aggregation.

6.2.1.2.40 UseCase root class

Specification of a set of actions performed by a system, which yields an observable result that is, typically, of value for one or more Actors or other stakeholders of the system

Table 140 shows all attributes of UseCase.

Table 140 – Attributes of UseCase_ExchangeModel::UseCase

name	type	description
name	String	a short name
identifier	String	The identification number (ID) of a use case is unique within a repository or project and serves for organization/administration of use cases.
nature	String	It classifies the main focus of the use case.
classification	String	At the international level the use case description might be generic enough to describe a use case in a more general way independently from the national or regional market design. But use cases might be used to describe regional or national specific circumstances like laws or even project specific details. If the use case reflects those circumstances, it should be characterized accordingly.
keywords	String	Keywords can be defined in order to support extended search functionalities within a use case repository. Multiple keywords should be provided as a comma-separated list.
prioritization	String	If considering a larger number of use cases, it might be interesting to cluster them according to priority. This prioritization might be different from country to country.
scope	String	The scope defines the limits of the use case.
mRID	String	master resource identifier

Table 141 shows all association ends of UseCase with other classes.

Table 141 – Association ends of UseCase_ExchangeModel::UseCase with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..*] ActorGrouping	ActorGrouping	A use case references several sets of Actors.
[0..1]	[0..*] Prerequisite	Condition	It describes what condition(s) should have been met before initiation of the use case, such as prior state of the Actors and activities.
[1..1]	[0..*] PrimaryActor	Ref_Actor	identification of the PrimaryActor linked to the use case
[0..1]	[0..*] BusinessCase	Ref_BusinessCase	list of business cases associated with the use case
[0..*]	[0..1]	UseCaseLibrary	This association end is an aggregation.
[0..1]	[0..*] Assumption	Assumption	set of assumptions related to the use case
[0..1]	[0..*] CustomInformation	CustomInformation	related additional information
[0..1]	[0..*] Drawing	Drawing	the list of associated drawings
[0..1]	[0..*] KeyPerformanceIndicator	KeyPerformanceIndicator	information about the KPI related to the use case
[0..1]	[0..1] Narrative	Narrative	the information about the narrative part of the use case
[0..1]	[0..*] RelatedObjective	Objective	list of objectives of the use case
[0..1]	[0..*] CommonTerm	Ref_CommonTerm	the associated common terms
[0..1]	[0..*] RelatedUseCase	Ref_UseCase	Known relations to other use cases can

[mult from]	[mult to] name	type	description
			be provided here if, for example, the use case is a more detailed one related to a high level use case, or it is an alternative to an existing use case.
[0..1]	[0..*] Reference	Reference	information about referenced documents
[0..1]	[0..*] Remark	Remark	remark associated with the use case
[0..1]	[0..*] Scenario	Scenario	reference to the scenarios belonging to the use case
[0..1]	[0..*] Version	Version	list of versions describing changes about the use case

6.2.1.2.41 UseCaseLibrary root class

Use cases of the repository.

Table 142 shows all attributes of UseCaseLibrary.

Table 142 – Attributes of UseCase_ExchangeModel::UseCaseLibrary

name	type	description
name	String	a short name

Table 143 shows all association ends of UseCaseLibrary with other classes.

Table 143 – Association ends of UseCase_ExchangeModel::UseCaseLibrary with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..1]	UseCaseRepository	This association end is an aggregation.
[0..1]	[0..*] UseCase	UseCase	list of use cases within the library

6.2.1.2.42 UseCaseRepository root class

Database, based on a given use cases template, for editing, maintenance and administration of use cases, Actors and requirements including their interrelations.

Table 144 shows all attributes of UseCaseRepository.

Table 144 – Attributes of UseCase_ExchangeModel::UseCaseRepository

name	type	description
name	String	a short name
description	String	description of the object
mRID	String	master resource identifier

Table 145 shows all association ends of UseCaseRepository with other classes.

Table 145 – Association ends of UseCase_ExchangeModel::UseCaseRepository with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..*] ActorLibrary	ActorLibrary	library of Actors
[0..1]	[0..1] AreaLibrary	AreaLibrary	the associated library of areas
[0..1]	[0..1] BusinessCaseLibrary	BusinessCaseLibrary	the library associated with the repository
[0..1]	[0..1] BusinessObjectLibrary	BusinessObjectLibrary	the library of business objects
[0..1]	[0..1] CommonTermLibrary	CommonTermLibrary	the associated library of common terms
[0..1]	[0..1] RequirementLibrary	RequirementLibrary	the associated library
[0..1]	[0..1] UseCaseLibrary	UseCaseLibrary	the associated library of use cases

6.2.1.2.43 Version root class

Version of the use case description.

Table 146 shows all attributes of Version.

Table 146 – Attributes of UseCase_ExchangeModel::Version

name	type	description
date	DateTime	date of creation of the version, in the format YYYY-MM-DD
changes	String	It represents the differences with respect to the previous version. Multiple changes are separated with paragraphs.
approvalStatus	String	information used within the standardization process
number	String	sequential number to identify the version of the document
mRID	String	master resource identifier

Table 147 shows all association ends of Version with other classes.

Table 147 – Association ends of UseCase_ExchangeModel::Version with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	UseCase	This association end is an aggregation.
[0..1]	[0..*] Author	Author	list of authors participating in the changes of this version

6.2.1.2.44 FurtherActorInformation root class

Individual or additional information that relates to the use case can be provided.

Table 148 shows all attributes of FurtherActorInformation.

Table 148 – Attributes of UseCase_ExchangeModel::FurtherActorInformation

name	type	description
description	String	description of the object
mRID	String	master resource identifier

Table 149 shows all association ends of FurtherActorInformation with other classes.

Table 149 – Association ends of UseCase_ExchangeModel::FurtherActorInformation with other classes

[mult from]	[mult to] name	type	description
[0..*]	[0..1]	ActorGrouping	This association end is an aggregation.
[0..1]	[0..*] Actor	Ref_Actor	information related to an Actor

6.2.2 Use case library XML syntactic format

Table 150 – Use case library XSD

<CODE BEGINS>

XSD content

```
<?xml version="1.0" encoding="utf-8"?>
<xsd:schema xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:UC="http://www.SYC.org/IEC62559/part3/UC/V01/Build08082016"
targetNamespace="http://www.SYC.org/IEC62559/part3/UC/V01/Build08082016">
	<xsd:complexType name="Activity">
		<xsd:annotation>
			<xsd:documentation>It may be subdivided into other activities or represent an elementary step of the
scenario.</xsd:documentation>
		</xsd:annotation>
		<xsd:sequence>
			<xsd:element name="BusinessObject" type="UC:Ref_BusinessObject" minOccurs="0"
maxOccurs="unbounded">
				<xsd:annotation>
					<xsd:documentation>This describes briefly the information to be exchanged between
the two actors – information producer and information receiver. It may list several information items in one step, comma
separated.</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="ChildActivity" type="UC:Activity" minOccurs="0" maxOccurs="unbounded">
				<xsd:annotation>
					<xsd:documentation>child activities</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="description" type="xsd:string" minOccurs="1" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>description of the object</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="Drawing" type="UC:Drawing" minOccurs="0" maxOccurs="unbounded">
				<xsd:annotation>
					<xsd:documentation>the list of associated drawing</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="event" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>It is the event that triggers the step. This might be completion of
previous steps (in that case the event is the comma-separated list of their numbers) or the condition to exit a
loop.</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="InformationProducer" type="UC:Ref_Actor" minOccurs="0"
maxOccurs="unbounded">
				<xsd:annotation>
					<xsd:documentation>This identifies the producer or source of the information. This
```

```
should be one of the actors defined in the repository.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="InformationReceiver" type="UC:Ref_Actor" minOccurs="0"
maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>This identifies the receiver of the information. This should be one
of the actors defined in the repository.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>master resource identifier</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>a short name</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="number" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>index of the activity</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="PrimaryActor" type="UC:Ref_Actor" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>reference to the PrimaryActor</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="Requirement" type="UC:Ref_Requirement" minOccurs="0"
maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>Several requirements may be listed in one step, comma
separated.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="service" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>This column identifies the nature of the information flow and the
originator of the information. Available options are CREATE, GET, CHANGE, DELETE, CANCEL, EXECUTE derived from
IEC 61968-100:2013.
Additionally, REPORT, TIMER and REPEAT are suggested.

	- CREATE means that an information object is to be created at the Producer.
	- GET (this is the default value if none is populated) means that the Receiver requests information from the Producer
(default).
	- CHANGE means that information is to be updated. Producer updates the Receiver’s information.
	- DELETE means that information is to be deleted. Producer deletes information from the Receiver.
	- CANCEL, CLOSE imply actions related to processes, such as the closure of a work order or the cancellation of a
control request.
	- EXECUTE is used when a complex transaction is being conveyed using a service, which potentially contains more
than one verb.
	- REPORT is used to represent transferral of unsolicited information or asynchronous information flows. Producer
provides information to the Receiver.
	- TIMER is used to represent a waiting period. When using the TIMER service, the Information Producer and
Information Receiver fields shall refer to the same actor.
	- REPEAT is used to indicate that a series of steps is repeated until a condition or trigger event. The condition is
specified as the text in the "Event" column for this row or step. Following the word REPEAT, the first and last step numbers of
the series to be repeated shall appear, in parenthesis, in the following form: REPEAT(X-Y) where X is the first step and Y is the
last step.

These common service definitions are related to automation/information or communication systems. In case the use case
template is applied in other domains, further services might be used and described.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="Drawing">
			<xsd:annotation>
				<xsd:documentation>For clarification, in general it is recommended to provide drawing(s) by hand, by
a graphic or as UML graphics (preferred in IEC 62559). The drawing should show interactions which identify the steps where
possible.</xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="drawingType" type="UC:DrawingClassification" minOccurs="1" maxOccurs="1">
					<xsd:annotation>
```

```
					<xsd:documentation>type of drawing</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>master resource identifier</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>a short name</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="URI" type="UC:Resource_String" minOccurs="1" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>resource Path to an image file
ex: http://www..../image.jpg
or
path to an XMI file with a relative diagram path from the root of the UML model. ex:
http://www.../Model.xmi#Package1/Package2/.../DiagName</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
		</xsd:sequence>
	</xsd:complexType>
	<xsd:simpleType name="DrawingClassification">
		<xsd:annotation>
			<xsd:documentation>possible types of drawing</xsd:documentation>
		</xsd:annotation>
		<xsd:restriction base="xsd:string">
			<xsd:enumeration value="domain overview">
				<xsd:annotation>
					<xsd:documentation>overview of relationships between domains and use
cases</xsd:documentation>
				</xsd:annotation>
			</xsd:enumeration>
			<xsd:enumeration value="use case overview">
				<xsd:annotation>
					<xsd:documentation>overview of use case short template
information</xsd:documentation>
				</xsd:annotation>
			</xsd:enumeration>
			<xsd:enumeration value="documentation">
				<xsd:annotation>
					<xsd:documentation>additional information about a use case</xsd:documentation>
				</xsd:annotation>
			</xsd:enumeration>
			<xsd:enumeration value="scenarios flowchart">
				<xsd:annotation>
					<xsd:documentation>flowchart of scenarios in a use case</xsd:documentation>
				</xsd:annotation>
			</xsd:enumeration>
			<xsd:enumeration value="scenario overview">
				<xsd:annotation>
					<xsd:documentation>overview of static information related to a
scenario</xsd:documentation>
				</xsd:annotation>
			</xsd:enumeration>
			<xsd:enumeration value="activities flowchart">
				<xsd:annotation>
					<xsd:documentation>flowchart of activities for describing a scenario or another activity
(in the case of recursive decomposition)</xsd:documentation>
				</xsd:annotation>
			</xsd:enumeration>
			<xsd:enumeration value="activity overview">
				<xsd:annotation>
					<xsd:documentation>static information related to an activity</xsd:documentation>
				</xsd:annotation>
			</xsd:enumeration>
			<xsd:enumeration value="business objects overview">
				<xsd:annotation>
					<xsd:documentation>overview of a set of objects used in information
exchanges</xsd:documentation>
				</xsd:annotation>
			</xsd:enumeration>
			<xsd:enumeration value="other">
				<xsd:annotation>
					<xsd:documentation>other kind of drawings</xsd:documentation>
```

```
                                </xsd:annotation>
                        </xsd:enumeration>
                        <xsd:enumeration value="role model">
                                <xsd:annotation>
                                        <xsd:documentation>overview of role and their relationships</xsd:documentation>
                                </xsd:annotation>
                        </xsd:enumeration>
                </xsd:restriction>
        </xsd:simpleType>
        <xsd:complexType name="Resource_String">
                <xsd:annotation>
                        <xsd:documentation>String Primitive type</xsd:documentation>
                </xsd:annotation>
                <xsd:simpleContent>
                        <xsd:extension base="UC:Resource_String_value_ValueType">
                                <xsd:attribute name="type" type="UC:ResourceType" use="required">
                                        <xsd:annotation>
                                                <xsd:documentation>supplementary attribute of the data
type</xsd:documentation>
                                        </xsd:annotation>
                                </xsd:attribute>
                        </xsd:extension>
                </xsd:simpleContent>
        </xsd:complexType>
        <xsd:simpleType name="Resource_String_value_ValueType">
                <xsd:restriction base="xsd:string"/>
        </xsd:simpleType>
        <xsd:simpleType name="ResourceType">
                <xsd:annotation>
                        <xsd:documentation>list of possible resources</xsd:documentation>
                </xsd:annotation>
                <xsd:restriction base="xsd:string">
                        <xsd:enumeration value="image">
                                <xsd:annotation>
                                        <xsd:documentation>any image kind for a resource</xsd:documentation>
                                </xsd:annotation>
                        </xsd:enumeration>
                        <xsd:enumeration value="UMLDiagram">
                                <xsd:annotation>
                                        <xsd:documentation>UML kind for a resource</xsd:documentation>
                                </xsd:annotation>
                        </xsd:enumeration>
                </xsd:restriction>
        </xsd:simpleType>
        <xsd:complexType name="Ref_Requirement">
                <xsd:annotation>
                        <xsd:documentation>requirement of a part of the use case</xsd:documentation>
                </xsd:annotation>
                <xsd:sequence>
                        <xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
                                <xsd:annotation>
                                        <xsd:documentation>master resource identifier</xsd:documentation>
                                </xsd:annotation>
                        </xsd:element>
                </xsd:sequence>
        </xsd:complexType>
        <xsd:complexType name="Ref_Actor">
                <xsd:annotation>
                        <xsd:documentation>It is the entity that communicates and interacts.
These actors can include people, software applications, systems, databases, and even the power system
itself.</xsd:documentation>
                </xsd:annotation>
                <xsd:sequence>
                        <xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
                                <xsd:annotation>
                                        <xsd:documentation>master resource identifier</xsd:documentation>
                                </xsd:annotation>
                        </xsd:element>
                </xsd:sequence>
        </xsd:complexType>
        <xsd:complexType name="Ref_BusinessObject">
                <xsd:annotation>
                        <xsd:documentation>business object exchanged in detailed activity steps</xsd:documentation>
                </xsd:annotation>
                <xsd:sequence>
                        <xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
                                <xsd:annotation>
```

```
						<xsd:documentation>master resource identifier</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="Actor">
			<xsd:annotation>
				<xsd:documentation>It is the entity that communicates and interacts.
These actors can include people, software applications, systems, databases, and even the power system
itself.</xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>description of the object</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>master resource identifier</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>a short name</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="type" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>type of the actor (business, system, ...)</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="ActorGrouping">
			<xsd:annotation>
				<xsd:documentation>group of actors used to organize an actor list</xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>description of the object</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="FurtherActorInformation" type="UC:FurtherActorInformation" minOccurs="0"
maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>In a context of one use case, an actor grouped into an
ActorGrouping may have one FurtherActorInformation.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>master resource identifier</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>a short name</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="FurtherActorInformation">
			<xsd:annotation>
				<xsd:documentation>Individual or additional information that relates to the use case can be
provided.</xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="Actor" type="UC:Ref_Actor" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>information related to an actor</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>description of the object</xsd:documentation>
```

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					</xsd:annotation>
				</xsd:element>
				<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>master resource identifier</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="ActorLibrary">
			<xsd:annotation>
				<xsd:documentation>actors of the repository</xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="Actor" type="UC:Actor" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>the actors listed in the library</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>a short name</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="Area">
			<xsd:annotation>
				<xsd:documentation>area of knowledge or activity characterized by a set of concepts and terminology
understood by the practitioners in that area</xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>description of the object</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>master resource identifier</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>a short name</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="Author">
			<xsd:annotation>
				<xsd:documentation>This field is used to document who has provided the current version. It can be a
person, organization or, for example, standardization committee like a technical committee (TC) or working group
(WG).</xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>master resource identifier</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>a short name</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="BusinessCase">
			<xsd:annotation>
				<xsd:documentation>It provides a description or reference with some rationale for the suggested use
case. Usually the business case is related to several use cases.</xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="Area" type="UC:Ref_Area" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>Use cases can be used in various areas (e.g. energy system).
```

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Within these areas different domains are used to define / determine a more specific subgrouping. One or more domains and
zones, comma separated, can be specified in the template.</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>description of the object</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>master resource identifier</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>a short name</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
        </xsd:sequence>
    </xsd:complexType>
    <xsd:complexType name="Ref_Area">
        <xsd:annotation>
            <xsd:documentation>area of knowledge or activity characterized by a set of concepts and terminology
understood by the practitioners in that area</xsd:documentation>
        </xsd:annotation>
        <xsd:sequence>
            <xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>master resource identifier</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
        </xsd:sequence>
    </xsd:complexType>
    <xsd:complexType name="BusinessCaseLibrary">
        <xsd:annotation>
            <xsd:documentation>business cases of the repository</xsd:documentation>
        </xsd:annotation>
        <xsd:sequence>
            <xsd:element name="BusinessCase" type="UC:BusinessCase" minOccurs="0"
maxOccurs="unbounded">
                <xsd:annotation>
                    <xsd:documentation>the list of business cases</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>a short name</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
        </xsd:sequence>
    </xsd:complexType>
    <xsd:complexType name="BusinessObject">
        <xsd:annotation>
            <xsd:documentation>business object exchanged in detailed activity steps</xsd:documentation>
        </xsd:annotation>
        <xsd:sequence>
            <xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>description of the object</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="identifier" type="xsd:string" minOccurs="1" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>identification used in references in the use case
description</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>master resource identifier</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>a short name</xsd:documentation>
                </xsd:annotation>
```

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						</xsd:element>
						<xsd:element name="Requirement" type="UC:Ref_Requirement" minOccurs="0"
maxOccurs="unbounded">
							<xsd:annotation>
								<xsd:documentation>It can be used to define requirements related to the information
and not to the step as in the step by step analysis.</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
					</xsd:sequence>
				</xsd:complexType>
				<xsd:complexType name="BusinessObjectLibrary">
					<xsd:annotation>
						<xsd:documentation>business objects which are exchanged in use cases</xsd:documentation>
					</xsd:annotation>
					<xsd:sequence>
						<xsd:element name="BusinessObject" type="UC:BusinessObject" minOccurs="0"
maxOccurs="unbounded">
							<xsd:annotation>
								<xsd:documentation>list of business objects within the library</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
						<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
							<xsd:annotation>
								<xsd:documentation>a short name</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
					</xsd:sequence>
				</xsd:complexType>
				<xsd:complexType name="CommonTerm">
					<xsd:annotation>
						<xsd:documentation>It represents a glossary term and its definition.</xsd:documentation>
					</xsd:annotation>
					<xsd:sequence>
						<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
							<xsd:annotation>
								<xsd:documentation>description of the object</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
						<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
							<xsd:annotation>
								<xsd:documentation>master resource identifier</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
						<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
							<xsd:annotation>
								<xsd:documentation>a short name</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
					</xsd:sequence>
				</xsd:complexType>
				<xsd:complexType name="CommonTermLibrary">
					<xsd:annotation>
						<xsd:documentation>common glossary defining terms for all use cases</xsd:documentation>
					</xsd:annotation>
					<xsd:sequence>
						<xsd:element name="CommonTerm" type="UC:CommonTerm" minOccurs="0"
maxOccurs="unbounded">
							<xsd:annotation>
								<xsd:documentation>list of common terms</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
						<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
							<xsd:annotation>
								<xsd:documentation>a short name</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
					</xsd:sequence>
				</xsd:complexType>
				<xsd:complexType name="Condition">
					<xsd:annotation>
						<xsd:documentation>It describes either a precondition or a postcondition.</xsd:documentation>
					</xsd:annotation>
					<xsd:sequence>
						<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
							<xsd:annotation>
								<xsd:documentation>description of the object</xsd:documentation>
							</xsd:annotation>
```

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					</xsd:element>
					<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>master resource identifier</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>a short name</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="number" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>index of the condition</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
				</xsd:sequence>
			</xsd:complexType>
			<xsd:complexType name="CustomInformation">
				<xsd:annotation>
					<xsd:documentation>It provides a flexible option to include miscellaneous custom, semi-structured
information which does not fit into other parts of the template.</xsd:documentation>
				</xsd:annotation>
				<xsd:sequence>
					<xsd:element name="key" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>unique key for identification purposes</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>master resource identifier</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="reference" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>This field refers to relevant template
section.</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="value" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>the corresponding information for the provided
key</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
				</xsd:sequence>
			</xsd:complexType>
			<xsd:complexType name="KeyPerformanceIndicator">
				<xsd:annotation>
					<xsd:documentation>Key Performance Indicator (KPI) related to the objectives of the use
case</xsd:documentation>
				</xsd:annotation>
				<xsd:sequence>
					<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>description of the object</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="identifier" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>unique identifier for the Key Performance Indicator
(KPI)</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>master resource identifier</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>a short name</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="Objective" type="UC:Ref_Objective" minOccurs="0" maxOccurs="unbounded">
						<xsd:annotation>
```

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<xsd:documentation>the related objective</xsd:documentation>
</xsd:annotation>
</xsd:element>
</xsd:sequence>
</xsd:complexType>
<xsd:complexType name="Ref_Objective">
<xsd:annotation>
<xsd:documentation>objective of the use case</xsd:documentation>
</xsd:annotation>
<xsd:sequence>
<xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>master resource identifier</xsd:documentation>
</xsd:annotation>
</xsd:element>
</xsd:sequence>
</xsd:complexType>
<xsd:complexType name="Narrative">
<xsd:annotation>
<xsd:documentation>It describes the narrative of the use case.</xsd:documentation>
</xsd:annotation>
<xsd:sequence>
<xsd:element name="completeDescription" type="xsd:string" minOccurs="0" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>It provides a complete narrative of the use case from a user's point of view, describing what occurs when, why, with what expectation, and under what conditions. This narrative should be written in plain text so that non-domain experts can understand it.
The length of the complete description can range from a few sentences to a few pages, depending on the complexity and/or newness of the use case. This description often helps the domain expert to reflect on the requirements for the use case before getting into the details in the next sections of the use case template.
The description may include drawings for explanation.</xsd:documentation>
</xsd:annotation>
</xsd:element>
<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>master resource identifier</xsd:documentation>
</xsd:annotation>
</xsd:element>
<xsd:element name="shortDescription" type="xsd:string" minOccurs="1" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>It is a short text intended to summarize the main idea as a service for the reader who is searching for a use case or looking for an overview.
Recommendation: This short description should have not more than 150 words.</xsd:documentation>
</xsd:annotation>
</xsd:element>
</xsd:sequence>
</xsd:complexType>
<xsd:complexType name="Objective">
<xsd:annotation>
<xsd:documentation>objective of the use case</xsd:documentation>
</xsd:annotation>
<xsd:sequence>
<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>description of the object</xsd:documentation>
</xsd:annotation>
</xsd:element>
<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>master resource identifier</xsd:documentation>
</xsd:annotation>
</xsd:element>
<xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>a short name</xsd:documentation>
</xsd:annotation>
</xsd:element>
</xsd:sequence>
</xsd:complexType>
<xsd:complexType name="Ref_BusinessCase">
<xsd:annotation>
<xsd:documentation>It provides a description or reference with some rationale for the suggested use case. Usually the business case is related to several use cases.</xsd:documentation>
</xsd:annotation>
<xsd:sequence>
<xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
<xsd:annotation>
```

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                        <xsd:documentation>master resource identifier</xsd:documentation>
                    </xsd:annotation>
                </xsd:element>
            </xsd:sequence>
        </xsd:complexType>
        <xsd:complexType name="Ref_CommonTerm">
            <xsd:annotation>
                <xsd:documentation>It represents a glossary term and its definition.</xsd:documentation>
            </xsd:annotation>
            <xsd:sequence>
                <xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
                    <xsd:annotation>
                        <xsd:documentation>master resource identifier</xsd:documentation>
                    </xsd:annotation>
                </xsd:element>
            </xsd:sequence>
        </xsd:complexType>
        <xsd:complexType name="Ref_UseCase">
            <xsd:annotation>
                <xsd:documentation>specification of a set of actions performed by a system, which yields an
observable result that is, typically, of value for one or more actors or other stakeholders of the system </xsd:documentation>
            </xsd:annotation>
            <xsd:sequence>
                <xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
                    <xsd:annotation>
                        <xsd:documentation>master resource identifier</xsd:documentation>
                    </xsd:annotation>
                </xsd:element>
            </xsd:sequence>
        </xsd:complexType>
        <xsd:complexType name="Reference">
            <xsd:annotation>
                <xsd:documentation>Any references that might restrict or affect the design, understanding, and
requirements of the use case shall be identified, including contracts, regulations, policies, financial considerations, engineering
constraints, pollution constraints, and other environmental quality issues.</xsd:documentation>
            </xsd:annotation>
            <xsd:sequence>
                <xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
                    <xsd:annotation>
                        <xsd:documentation>description of the object</xsd:documentation>
                    </xsd:annotation>
                </xsd:element>
                <xsd:element name="impact" type="xsd:string" minOccurs="0" maxOccurs="1">
                    <xsd:annotation>
                        <xsd:documentation>This information describes the nature of the influence of the
document on the use case.</xsd:documentation>
                    </xsd:annotation>
                </xsd:element>
                <xsd:element name="link" type="xsd:string" minOccurs="0" maxOccurs="1">
                    <xsd:annotation>
                        <xsd:documentation>If available, a public link to the document can be
provided.</xsd:documentation>
                    </xsd:annotation>
                </xsd:element>
                <xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
                    <xsd:annotation>
                        <xsd:documentation>master resource identifier</xsd:documentation>
                    </xsd:annotation>
                </xsd:element>
                <xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
                    <xsd:annotation>
                        <xsd:documentation>a short name</xsd:documentation>
                    </xsd:annotation>
                </xsd:element>
                <xsd:element name="number" type="xsd:string" minOccurs="0" maxOccurs="1">
                    <xsd:annotation>
                        <xsd:documentation>index of the reference</xsd:documentation>
                    </xsd:annotation>
                </xsd:element>
                <xsd:element name="originatorOrganization" type="xsd:string" minOccurs="1" maxOccurs="1">
                    <xsd:annotation>
                        <xsd:documentation>This describes the name of the organization publishing the
document.</xsd:documentation>
                    </xsd:annotation>
                </xsd:element>
                <xsd:element name="status" type="xsd:string" minOccurs="0" maxOccurs="1">
                    <xsd:annotation>
```

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					<xsd:documentation>status of the referenced document</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="type" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>There are different reference types (e.g. standard, regulation, contract, others like publications).</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
		</xsd:sequence>
	</xsd:complexType>
	<xsd:complexType name="Remark">
		<xsd:annotation>
			<xsd:documentation>It is used for further comments which are not considered elsewhere.</xsd:documentation>
		</xsd:annotation>
		<xsd:sequence>
			<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>description of the object</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>master resource identifier</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
		</xsd:sequence>
	</xsd:complexType>
	<xsd:complexType name="Requirement">
		<xsd:annotation>
			<xsd:documentation>requirement of a part of the use case</xsd:documentation>
		</xsd:annotation>
		<xsd:sequence>
			<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>description of the object</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="identifier" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>identification used in references</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>master resource identifier</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>a short name</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
		</xsd:sequence>
	</xsd:complexType>
	<xsd:complexType name="RequirementCategory">
		<xsd:annotation>
			<xsd:documentation>Requirements can be sorted in categories. For complex categories the category name can be "dot-delimited".</xsd:documentation>
		</xsd:annotation>
		<xsd:sequence>
			<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>description of the object</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="identifier" type="xsd:string" minOccurs="1" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>identification used in references</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>master resource identifier</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
```

```
				<xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>a short name</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="Requirement" type="UC:Requirement" minOccurs="0"
maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>information about the requirement</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="RequirementLibrary">
			<xsd:annotation>
				<xsd:documentation>requirements of the repository</xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>a short name</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="RequirementCategory" type="UC:RequirementCategory" minOccurs="0"
maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>the category for the requirement</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="Scenario">
			<xsd:annotation>
				<xsd:documentation>a possible sequence of interactions</xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="Activity" type="UC:Activity" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>list of activities for the scenario</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>description of the object</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="Drawing" type="UC:Drawing" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>the list of associated drawing</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>master resource identifier</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>a short name</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="number" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>index of the scenario</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="Precondition" type="UC:Condition" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>It describes which condition(s) should have been met before this
scenario happens.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="Postcondition" type="UC:Condition" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>It describes which condition(s) should prevail after this scenario
happens. The post conditions may also define "success" or "failure" conditions for each use case. </xsd:documentation>
					</xsd:annotation>
```

```
				</xsd:element>
				<xsd:element name="PrimaryActor" type="UC:Ref_Actor" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>It describes which actor triggers this
scenario.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="Requirement" type="UC:Ref_Requirement" minOccurs="0"
maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>information about the requirement</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="TriggeringEvent" type="UC:TriggeringEvent" minOccurs="0"
maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>list of triggering events for the scenario</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="TriggeringEvent">
			<xsd:annotation>
				<xsd:documentation>It describes which event triggers a scenario.</xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>description of the object</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>master resource identifier</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>a short name</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="UseCase">
			<xsd:annotation>
				<xsd:documentation>specification of a set of actions performed by a system, which yields an
observable result that is, typically, of value for one or more actors or other stakeholders of the system </xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="ActorGrouping" type="UC:ActorGrouping" minOccurs="0"
maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>A use case references several sets of
actors.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="Assumption" type="UC:Assumption" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>set of assumptions related to the use case</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="BusinessCase" type="UC:Ref_BusinessCase" minOccurs="0"
maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>list of business cases associated with the use
case</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="classification" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>At the international level the use case description might be
generic enough to describe a use case in a more general way independently from the national or regional market design. But
use cases might be used to describe regional or national specific circumstances like laws or even project specific details. If the
use case reflects those circumstances, it should be characterized accordingly.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="CommonTerm" type="UC:Ref_CommonTerm" minOccurs="0"
```

```
maxOccurs="unbounded">
                <xsd:annotation>
                    <xsd:documentation>the associated common terms</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="CustomInformation" type="UC:CustomInformation" minOccurs="0"
maxOccurs="unbounded">
                <xsd:annotation>
                    <xsd:documentation>related additional information</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="Drawing" type="UC:Drawing" minOccurs="0" maxOccurs="unbounded">
                <xsd:annotation>
                    <xsd:documentation>the list of associated drawings</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="identifier" type="xsd:string" minOccurs="1" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>The identification number (ID) of a use case is unique within a
repository or project and serves for organization/administration of use cases.</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="KeyPerformanceIndicator" type="UC:KeyPerformanceIndicator" minOccurs="0"
maxOccurs="unbounded">
                <xsd:annotation>
                    <xsd:documentation>information about the KPI related to the use
case</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="keywords" type="xsd:string" minOccurs="0" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>Keywords can be defined in order to support extended search
functionalities within a use case repository. Multiple keywords should be provided as a comma-separated
list.</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="levelOfDepth" type="xsd:string" minOccurs="0" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>Use cases can be described on different
levels.</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>master resource identifier</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>a short name</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="Narrative" type="UC:Narrative" minOccurs="0" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>the information about the narrative part of the use
case</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="nature" type="xsd:string" minOccurs="0" maxOccurs="1">
                <xsd:annotation>
                    <xsd:documentation>It classifies the main focus of the use case.</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="Prerequisite" type="UC:Condition" minOccurs="0" maxOccurs="unbounded">
                <xsd:annotation>
                    <xsd:documentation>It describes what condition(s) should have been met before
initiation of the use case, such as prior state of the actors and activities.</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="PrimaryActor" type="UC:Ref_Actor" minOccurs="0" maxOccurs="unbounded">
                <xsd:annotation>
                    <xsd:documentation>identification of the PrimaryActor linked to the use
case</xsd:documentation>
                </xsd:annotation>
            </xsd:element>
            <xsd:element name="prioritization" type="xsd:string" minOccurs="0" maxOccurs="1">
                <xsd:annotation>
```

```
						<xsd:documentation>If considering a larger number of use cases, it might be
interesting to cluster them according to priority.
This prioritization might be different from country to country.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="Reference" type="UC:Reference" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>information about referenced documents</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="RelatedObjective" type="UC:Objective" minOccurs="0"
maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>list of objectives of the use case</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="RelatedUseCase" type="UC:Ref_UseCase" minOccurs="0"
maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>Known relations to other use cases can be provided here if, for
example, the use case is a more detailed one related to a high level use case, or it is an alternative to an existing use
case.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="Remark" type="UC:Remark" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>remark associated with the use case</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="Scenario" type="UC:Scenario" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>reference to the scenarios belonging to the use
case</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="scope" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>The scope defines the limits of the use
case.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="Version" type="UC:Version" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>list of versions describing changes about the use
case</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
			</xsd:sequence>
		</xsd:complexType>
		<xsd:complexType name="Version">
			<xsd:annotation>
				<xsd:documentation>version of the use case description</xsd:documentation>
			</xsd:annotation>
			<xsd:sequence>
				<xsd:element name="approvalStatus" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>information used within the standardization
process</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="Author" type="UC:Author" minOccurs="0" maxOccurs="unbounded">
					<xsd:annotation>
						<xsd:documentation>list of authors participating in the changes of this
version</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="changes" type="xsd:string" minOccurs="0" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>It represents the differences with respect to the previous version.
Multiple changes are separated with paragraphs.</xsd:documentation>
					</xsd:annotation>
				</xsd:element>
				<xsd:element name="date" type="xsd:dateTime" minOccurs="1" maxOccurs="1">
					<xsd:annotation>
						<xsd:documentation>date of creation of the version, in the format YYYY-MM-
DD</xsd:documentation>
					</xsd:annotation>
```

```
			</xsd:element>
			<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>master resource identifier</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="number" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>sequential number to identify the version of the
document</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
		</xsd:sequence>
	</xsd:complexType>
	<xsd:complexType name="UseCaseLibrary">
		<xsd:annotation>
			<xsd:documentation>use cases of the repository</xsd:documentation>
		</xsd:annotation>
		<xsd:sequence>
			<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>a short name</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="UseCase" type="UC:UseCase" minOccurs="0" maxOccurs="unbounded">
				<xsd:annotation>
					<xsd:documentation>list of use cases within the library</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
		</xsd:sequence>
	</xsd:complexType>
	<xsd:complexType name="UseCaseRepository_Type">
		<xsd:annotation>
			<xsd:documentation>database, based on a given use cases template, for editing, maintenance and
administration of use cases, actors and requirements including their interrelations</xsd:documentation>
		</xsd:annotation>
		<xsd:sequence>
			<xsd:element name="ActorLibrary" type="UC:ActorLibrary" minOccurs="0" maxOccurs="unbounded">
				<xsd:annotation>
					<xsd:documentation>library of actors</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="AreaLibrary" type="UC:AreaLibrary" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>the associated library of areas</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="BusinessCaseLibrary" type="UC:BusinessCaseLibrary" minOccurs="0"
maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>the library associated with the repository</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="BusinessObjectLibrary" type="UC:BusinessObjectLibrary" minOccurs="0"
maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>the library of business objects</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="CommonTermLibrary" type="UC:CommonTermLibrary" minOccurs="0"
maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>the associated library of common terms</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>description of the object</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>master resource identifier</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
				<xsd:annotation>
```

```
<xsd:documentation>a short name</xsd:documentation>
</xsd:annotation>
</xsd:element>
<xsd:element name="RequirementLibrary" type="UC:RequirementLibrary" minOccurs="0"
maxOccurs="1">
<xsd:annotation>
<xsd:documentation>the associated library</xsd:documentation>
</xsd:annotation>
</xsd:element>
<xsd:element name="UseCaseLibrary" type="UC:UseCaseLibrary" minOccurs="0" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>the associated library of use cases</xsd:documentation>
</xsd:annotation>
</xsd:element>
</xsd:sequence>
</xsd:complexType>
<xsd:element name="UseCaseRepository" type="UC:UseCaseRepository_Type"/>
<xsd:complexType name="AreaLibrary">
<xsd:annotation>
<xsd:documentation>areas of the repository</xsd:documentation>
</xsd:annotation>
<xsd:sequence>
<xsd:element name="Area" type="UC:Area" minOccurs="0" maxOccurs="unbounded">
<xsd:annotation>
<xsd:documentation>the list of areas within the library</xsd:documentation>
</xsd:annotation>
</xsd:element>
<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>a short name</xsd:documentation>
</xsd:annotation>
</xsd:element>
</xsd:sequence>
</xsd:complexType>
<xsd:complexType name="Assumption">
<xsd:annotation>
<xsd:documentation>It may be used to define a further general assumption for a use case, such as
which systems already exist, which contractual relations exist, and which configurations of systems are probably in place. Initial
states of information exchanged shall also be identified.</xsd:documentation>
</xsd:annotation>
<xsd:sequence>
<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>description of the object</xsd:documentation>
</xsd:annotation>
</xsd:element>
<xsd:element name="mRID" type="xsd:string" minOccurs="0" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>master resource identifier</xsd:documentation>
</xsd:annotation>
</xsd:element>
<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>a short name</xsd:documentation>
</xsd:annotation>
</xsd:element>
<xsd:element name="number" type="xsd:string" minOccurs="0" maxOccurs="1">
<xsd:annotation>
<xsd:documentation>index of the assumption</xsd:documentation>
</xsd:annotation>
</xsd:element>
</xsd:sequence>
</xsd:complexType>
</xsd:schema>
```

<CODE ENDS>

6.3 Actor library exchange Format

6.3.1 UML actor library exchange model (UML Contextual Model)

6.3.1.1 General

The Actor library exchange model is described through a UML Contextual Model. The XSD content is generated from this model.

6.3.1.2 Model package ActorLibrary_ExchangeModel

6.3.1.2.1 General

Figure 23 shows class diagram ActorLibrary.

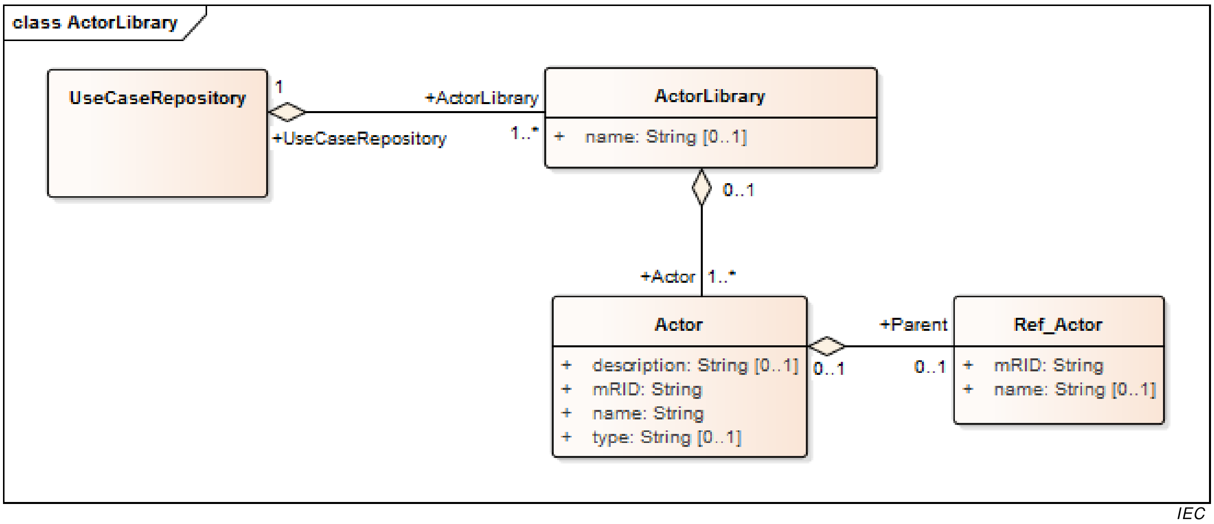

Figure 23 – Class diagram ActorLibrary_ExchangeModel::ActorLibrary

Figure 24 shows package diagram ActorLibrary.

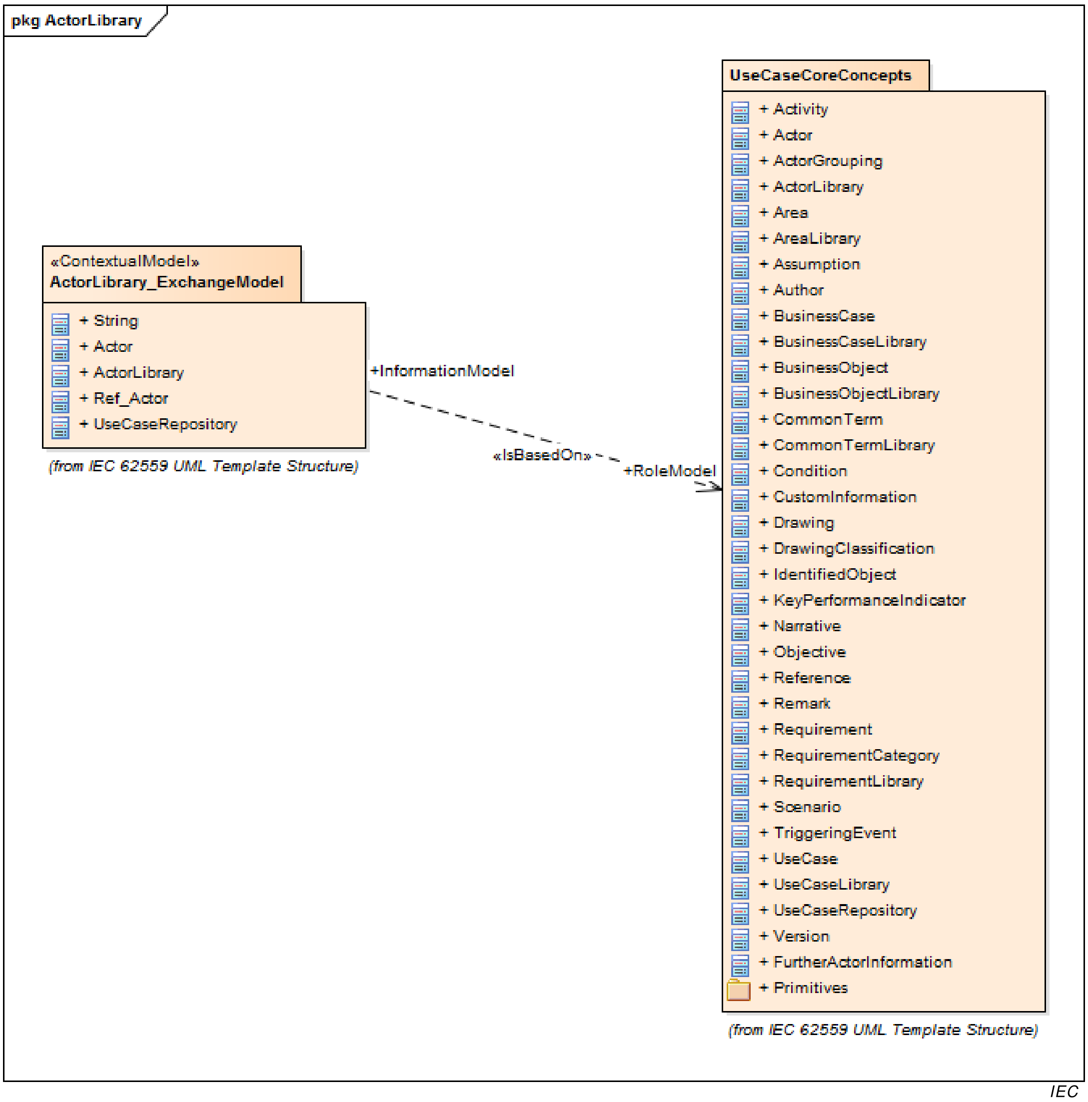

Figure 24 – Package diagram ActorLibrary_ExchangeModel::ActorLibrary

6.3.1.2.2 String primitive

String Primitive type

6.3.1.2.3 Actor root class

It is the entity that communicates and interacts.

These Actors can include people, software applications, systems, databases, and even the power system itself.

Table 151 shows all attributes of Actor.

Table 151 – Attributes of ActorLibrary_ExchangeModel::Actor

name	type	description
type	String	type of the Actor (business, system, ...)
description	String	description of the object
mRID	String	master resource identifier
name	String	a short name

Table 152 shows all association ends of Actor with other classes.

Table 152 – Association ends of ActorLibrary_ExchangeModel::Actor with other classes

[mult from]	[mult to] name	type	description
[1..*]	[0..1]	ActorLibrary	This association end is an aggregation.
[0..1]	[0..1] Parent	Ref_Actor	the generalized Actor

6.3.1.2.4 ActorLibrary root class

Actors of the repository.

Table 153 shows all attributes of ActorLibrary.

Table 153 – Attributes of ActorLibrary_ExchangeModel::ActorLibrary

name	type	description
name	String	a short name

Table 154 shows all association ends of ActorLibrary with other classes.

Table 154 – Association ends of ActorLibrary_ExchangeModel::ActorLibrary with other classes

[mult from]	[mult to] name	type	description
[0..1]	[1..*] Actor	Actor	the Actors listed in the library
[1..*]	[1..1] UseCaseRepository	UseCaseRepository	This association end is an aggregation.

6.3.1.2.5 Ref_Actor root class

It is the entity that communicates and interacts.

These Actors can include people, software applications, systems, databases, and even the power system itself.

Table 155 shows all attributes of Ref_Actor.

Table 155 – Attributes of ActorLibrary_ExchangeModel::Ref_Actor

name	type	description
mRID	String	master resource identifier
name	String	a short name

Table 156 shows all association ends of Ref_Actor with other classes.

Table 156 – Association ends of ActorLibrary_ExchangeModel::Ref_Actor with other classes

[mult from]	[mult to] name	type	description
[0..1]	[0..1]	Actor	This association end is an aggregation.

6.3.1.2.6 UseCaseRepository root class

Database, based on a given use cases template, for editing, maintenance and administration of use cases, Actors and requirements including their interrelations.

Table 157 shows all association ends of UseCaseRepository with other classes.

Table 157 – Association ends of ActorLibrary_ExchangeModel:: UseCaseRepository with other classes

[mult from]	[mult to] name	type	description
[1..1]	[1..*] ActorLibrary	ActorLibrary	library of Actors

6.3.2 Actor library XML syntactic format

Table 158 – Actor library XSD

<CODE BEGINS>

XSD content

```
<?xml version="1.0" encoding="utf-8"?>
<xsd:schema xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:AL="http://www.SYC.org/IEC62559/part3/AL/V01/Build08082016"
targetNamespace="http://www.SYC.org/IEC62559/part3/AL/V01/Build08082016">
	<xsd:complexType name="Actor">
		<xsd:annotation>
			<xsd:documentation>It is the entity that communicates and interacts.
These actors can include people, software applications, systems, databases, and even the power system
itself.</xsd:documentation>
		</xsd:annotation>
		<xsd:sequence>
			<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>description of the object</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>master resource identifier</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>a short name</xsd:documentation>
				</xsd:annotation>
```

```
						</xsd:element>
						<xsd:element name="Parent" type="AL:Ref_Actor" minOccurs="0" maxOccurs="1">
							<xsd:annotation>
								<xsd:documentation>the generalized actor</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
						<xsd:element name="type" type="xsd:string" minOccurs="0" maxOccurs="1">
							<xsd:annotation>
								<xsd:documentation>type of the actor (business, system, ...)</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
					</xsd:sequence>
				</xsd:complexType>
				<xsd:complexType name="Ref_Actor">
					<xsd:annotation>
						<xsd:documentation>It is the entity that communicates and interacts.
These actors can include people, software applications, systems, databases, and even the power system
itself.</xsd:documentation>
					</xsd:annotation>
					<xsd:sequence>
						<xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
							<xsd:annotation>
								<xsd:documentation>master resource identifier</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
						<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
							<xsd:annotation>
								<xsd:documentation>a short name</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
					</xsd:sequence>
				</xsd:complexType>
				<xsd:complexType name="ActorLibrary">
					<xsd:annotation>
						<xsd:documentation>actors of the repository</xsd:documentation>
					</xsd:annotation>
					<xsd:sequence>
						<xsd:element name="Actor" type="AL:Actor" minOccurs="1" maxOccurs="unbounded">
							<xsd:annotation>
								<xsd:documentation>the actors listed in the library</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
						<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
							<xsd:annotation>
								<xsd:documentation>a short name</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
					</xsd:sequence>
				</xsd:complexType>
				<xsd:complexType name="UseCaseRepository_Type">
					<xsd:annotation>
						<xsd:documentation>database, based on a given use cases template, for editing, maintenance and
administration of use cases, actors and requirements including their interrelations</xsd:documentation>
					</xsd:annotation>
					<xsd:sequence>
						<xsd:element name="ActorLibrary" type="AL:ActorLibrary" minOccurs="1" maxOccurs="unbounded">
							<xsd:annotation>
								<xsd:documentation>library of actors</xsd:documentation>
							</xsd:annotation>
						</xsd:element>
					</xsd:sequence>
				</xsd:complexType>
				<xsd:element name="UseCaseRepository" type="AL:UseCaseRepository_Type"/>
</xsd:schema>
```

<CODE ENDS>

6.4 Requirement library exchange Format

6.4.1 UML requirement library exchange model (UML Contextual Model)

6.4.1.1 General

The requirement library exchange model is described through a UML Contextual Model. The XSD content is generated from this model.

6.4.1.2 Model package RequirementLibrary_ExchangeModel

6.4.1.2.1 General

Figure 25 shows class diagram RequirementLibrary.

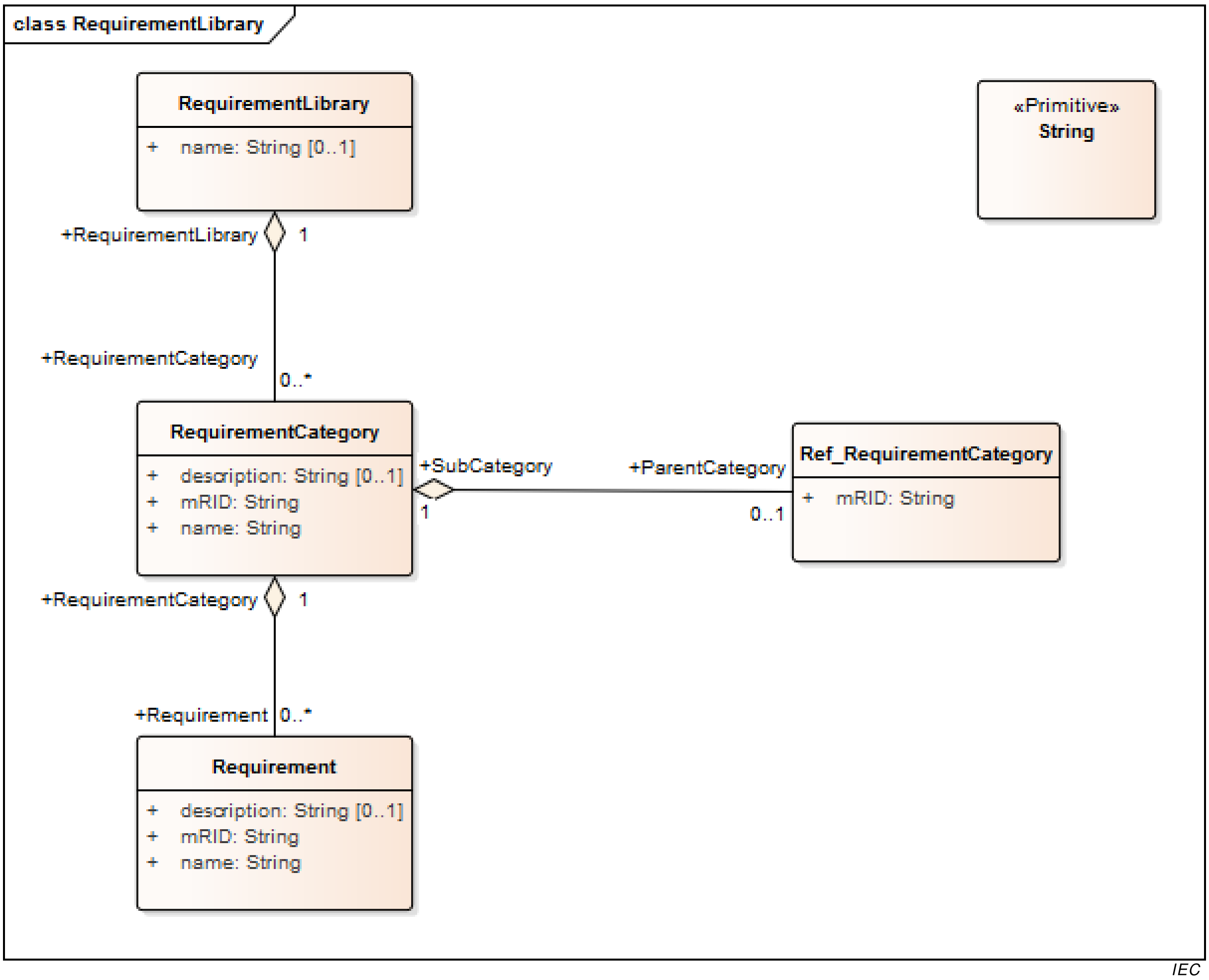

Figure 25 – Class diagram RequirementLibrary_ExchangeModel::RequirementLibrary

Figure 26 shows package diagram RequirementLibrary.

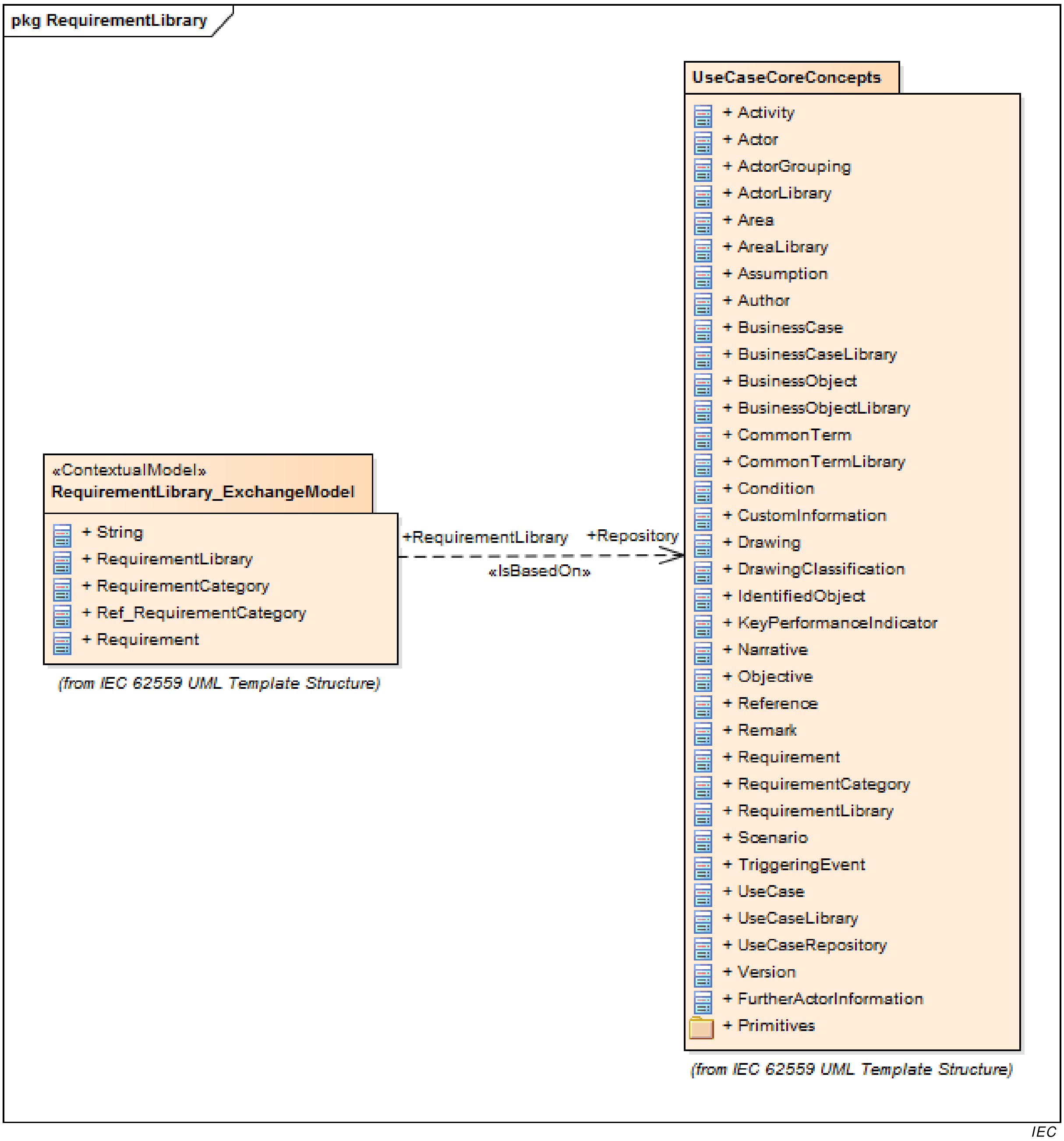

Figure 26 – Package diagram RequirementLibrary_ExchangeModel::RequirementLibrary

6.4.1.2.2 String primitive

String Primitive type

6.4.1.2.3 RequirementLibrary root class

Requirements of the repository.

Table 159 shows all attributes of RequirementLibrary.

Table 159 – Attributes of RequirementLibrary_ExchangeModel::RequirementLibrary

name	type	description
name	String	a short name

Table 160 shows all association ends of RequirementLibrary with other classes.

Table 160 – Association ends of RequirementLibrary_ExchangeModel::RequirementLibrary with other classes

[mult from]	[mult to] name	type	description
[1..1]	[0..*] RequirementCategory	RequirementCategory	the category for the requirement

6.4.1.2.4 RequirementCategory root class

Requirements can be sorted in categories. For complex categories the category name can be "dot-delimited".

Table 161 shows all attributes of RequirementCategory.

Table 161 – Attributes of RequirementLibrary_ExchangeModel::RequirementCategory

name	type	description
description	String	description of the object
mRID	String	master resource identifier
name	String	a short name

Table 162 shows all association ends of RequirementCategory with other classes.

Table 162 – Association ends of RequirementLibrary_ExchangeModel::RequirementCategory with other classes

[mult from]	[mult to] name	type	description
[0..*]	[1..1] RequirementLibrary	RequirementLibrary	This association end is an aggregation.
[1..1]	[0..1] ParentCategory	Ref_RequirementCategory	the reference to the parent category of requirement
[1..1]	[0..*] Requirement	Requirement	information about the requirement

6.4.1.2.5 Ref_RequirementCategory root class

Requirements can be sorted in categories. For complex categories the category name can be "dot-delimited".

Table 163 shows all attributes of Ref_RequirementCategory.

Table 163 – Attributes of RequirementLibrary_ExchangeModel::Ref_RequirementCategory

name	type	description
mRID	String	master resource identifier

Table 164 shows all association ends of Ref_RequirementCategory with other classes.

Table 164 – Association ends of RequirementLibrary_ExchangeModel:: Ref_RequirementCategory with other classes

[mult from]	[mult to] name	type	description
[0..1]	[1..1] SubCategory	RequirementCategory	This association end is an aggregation.

6.4.1.2.6 Requirement root class

Requirement of a part of the use case.

Table 165 shows all attributes of Requirement.

Table 165 – Attributes of RequirementLibrary_ExchangeModel::Requirement

name	type	description
description	String	description of the object
mRID	String	master resource identifier
name	String	a short name

Table 166 shows all association ends of Requirement with other classes.

Table 166 – Association ends of RequirementLibrary_ExchangeModel::Requirement with other classes

[mult from]	[mult to] name	type	description
[0..*]	[1..1] RequirementCategory	RequirementCategory	This association end is an aggregation.

6.4.2 Requirement library XML syntactic format

Table 167 – Requirement library XSD

<CODE BEGINS>

XSD content

```
<?xml version="1.0" encoding="utf-8"?>
<xsd:schema xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:RL="http://www.SYC.org/IEC62559/part3/RL/V01/Build08082016"
targetNamespace="http://www.SYC.org/IEC62559/part3/RL/V01/Build08082016">
	<xsd:complexType name="RequirementLibrary_Type">
		<xsd:annotation>
			<xsd:documentation>requirements of the repository</xsd:documentation>
		</xsd:annotation>
		<xsd:sequence>
			<xsd:element name="name" type="xsd:string" minOccurs="0" maxOccurs="1">
				<xsd:annotation>
					<xsd:documentation>a short name</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
			<xsd:element name="RequirementCategory" type="RL:RequirementCategory" minOccurs="0"
maxOccurs="unbounded">
				<xsd:annotation>
					<xsd:documentation>the category for the requirement</xsd:documentation>
				</xsd:annotation>
			</xsd:element>
		</xsd:sequence>
	</xsd:complexType>
	<xsd:complexType name="RequirementCategory">
		<xsd:annotation>
			<xsd:documentation>Requirements can be sorted in categories. For complex categories the category
```

```
name can be "dot-delimited".</xsd:documentation>
				</xsd:annotation>
				<xsd:sequence>
					<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>description of the object</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>master resource identifier</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>a short name</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="ParentCategory" type="RL:Ref_RequirementCategory" minOccurs="0"
maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>the reference to the parent category of
requirement</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="Requirement" type="RL:Requirement" minOccurs="0"
maxOccurs="unbounded">
						<xsd:annotation>
							<xsd:documentation>information about the requirement</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
				</xsd:sequence>
			</xsd:complexType>
			<xsd:complexType name="Ref_RequirementCategory">
				<xsd:annotation>
					<xsd:documentation>Requirements can be sorted in categories. For complex categories the category
name can be "dot-delimited".</xsd:documentation>
				</xsd:annotation>
				<xsd:sequence>
					<xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>master resource identifier</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
				</xsd:sequence>
			</xsd:complexType>
			<xsd:complexType name="Requirement">
				<xsd:annotation>
					<xsd:documentation>requirement of a part of the use case</xsd:documentation>
				</xsd:annotation>
				<xsd:sequence>
					<xsd:element name="description" type="xsd:string" minOccurs="0" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>description of the object</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="mRID" type="xsd:string" minOccurs="1" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>master resource identifier</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
					<xsd:element name="name" type="xsd:string" minOccurs="1" maxOccurs="1">
						<xsd:annotation>
							<xsd:documentation>a short name</xsd:documentation>
						</xsd:annotation>
					</xsd:element>
				</xsd:sequence>
			</xsd:complexType>
			<xsd:element name="RequirementLibrary" type="RL:RequirementLibrary_Type"/>
</xsd:schema>
```

<CODE ENDS>

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